Statistical Inference

Series 12 • Graded Exercise

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Note: This document contains only the graded problem, Exercise 3. All power functions are computed exactly *(no simulation). The figures are made by a Python script (`engine.py`, `make_figs.py`) supplied with this file.*

Exercise 3

For each of the following examples, develop the likelihood ratio test, Wald's test and Rao's score test for testing $H_0 : \theta = \theta_0$ against $H_{\text{left}} : \theta < \theta_0$, $H_{\text{right}} : \theta > \theta_0$, and $H_{\text{both}} : \theta \neq \theta_0$. Clearly specify the test statistic and the cut-off value. In each case, for a fixed sample size n, draw the power functions of the three tests on the same graph (fixing some specific value of θ_0), along with the power function of the UMP test (for one-sided alternatives) or the UMPU test (for two-sided alternatives) derived in the previous exercise. Repeat each setup for $n = 5, 10, 20, 40, 80, 160$. Based on these plots, comment on the relative merits/demerits of the three tests, and compare them with the optimal (most powerful) testing procedures. Does your conclusion change if the value of θ_0 is changed?

(a) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Binomial}(p)$, $0 \leq p \leq 1$, $\theta = p$.

(b) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Poisson}(\lambda)$, $\lambda > 0$, $\theta = \lambda$.

(c) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Geometric}(p)$, $0 \leq p \leq 1$, $\theta = p$.

(d) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$.

(e) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Normal}(\mu, 1)$, $\mu \in \mathbb{R}$, $\theta = \mu$.

(f) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Normal}(0, \sigma^2)$, $\sigma > 0$.

(g) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exponential}(\theta)$, $\theta > 0$.

(h) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Laplace}(0, \sigma)$, $\sigma > 0$, $\theta = \sigma$.

(i) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Normal}(\theta, \theta)$, $\theta > 0$.

Solution:

Set-up

Throughout, $\alpha = 0.05$, $z_{0.95} = 1.645$ and $z_{0.975} = 1.960$. Let $\ell(\theta)$ be the log-likelihood, $U_n(\theta) = \ell'(\theta)$ the score, $I(\theta)$ the Fisher information of one observation ($I_n = nI$) and $\hat{\theta}$

the MLE. The three signed statistics, each approximately $N(0,1)$ under H_0, are

$$\text{LRT:}\quad R_n = \operatorname{sgn}(\hat\theta-\theta_0)\sqrt{-2\log\Lambda_n}, \qquad -2\log\Lambda_n = 2\left[\ell(\hat\theta)-\ell(\theta_0)\right]$$

$$\text{Wald:}\quad W_n = (\hat\theta-\theta_0)\sqrt{I_n(\hat\theta)}$$

$$\text{Score:}\quad S_n = \frac{U_n(\theta_0)}{\sqrt{I_n(\theta_0)}}$$

Wald uses the information at $\hat\theta$; the score test uses it at θ_0 and needs no MLE. For $T_n \in \{R_n, W_n, S_n\}$ the cut-offs are

$$\text{reject } H_{\text{right}} \text{ if } T_n > z_{0.95}, \qquad \text{reject } H_{\text{left}} \text{ if } T_n < -z_{0.95}, \qquad \text{reject } H_{\text{both}} \text{ if } |T_n| > z_{0.975}$$

(for H_{both} this is $T_n^2 > \chi^2_{1,0.95} = 3.841$).

Plots. All three statistics depend on the data only through the sufficient statistic, so each power function is computed exactly (by summing the pmf, or by using the exact distribution of $\hat\theta$). The UMP/UMPU tests are the ones from Exercise 2. In every plot: thick grey = UMP/UMPU, solid blue = LRT, dashed orange = Wald, dotted red = score. Dash-dotted grey lines mark $\alpha = 0.05$ and θ_0. A missing curve lies exactly under another one. The horizontal axis is $\theta_0 \pm 4.5/\sqrt{nI(\theta_0)}$, clipped to the parameter space. Each part has a table of exact sizes; the UMP/UMPU tests have size exactly 0.05 (randomized when the model is discrete).

(a) $T = \sum X_i \sim \text{Bin}(n,p)$ is sufficient, and

$$\ell(p) = T\log p + (n-T)\log(1-p), \qquad U_n(p) = \frac{T-np}{p(1-p)}, \qquad I(p) = \frac{1}{p(1-p)}, \qquad \hat p = \frac{T}{n}$$

Test statistics (p_0 is the null value):

$$R_n = \operatorname{sgn}(\hat p - p_0)\sqrt{2n\left[\hat p\log\frac{\hat p}{p_0} + (1-\hat p)\log\frac{1-\hat p}{1-p_0}\right]}$$

$$W_n = \frac{\sqrt{n}(\hat p - p_0)}{\sqrt{\hat p(1-\hat p)}}, \qquad S_n = \frac{\sqrt{n}(\hat p - p_0)}{\sqrt{p_0(1-p_0)}}$$

with $0\log 0 = 0$, and $W_n = \pm\infty$ (so we reject) if $\hat p \in \{0,1\}$.

Optimal tests. The pmf is $\propto \exp\{a(p)T\}$ with $a(p) = \log\frac{p}{1-p}$ increasing, so there is MLR in T. The UMP test for H_{right} rejects if $T > c$, and with probability γ if $T = c$, where $P_{p_0}(T > c) + \gamma P_{p_0}(T = c) = \alpha$ (similarly for H_{left} with $T < c$). The UMPU test rejects if $T < c_1$ or $T > c_2$, with probabilities γ_1, γ_2 at the boundaries, where $E_{p_0}[\phi] = \alpha$ and $E_{p_0}[T\phi] = \alpha n p_0$.

Observations.

- *Sizes are erratic.* T is discrete and the cut-offs are asymptotic, so sizes jump around α. For $p_0 = 0.5$, $n = 5$: LRT and score have size 0.031, Wald has 0.188 (it rejects H_{left} whenever $T \le 1$). For $n = 10$ and H_{both}: LRT and Wald 0.109, score 0.021. Only for $n \ge 80$ do all sizes lie in $[0.046, 0.057]$.

- *Power.* A coloured curve above the grey one near p_0 only reflects a larger size, not a better test. For $n \geq 80$ all three tests are close to the UMP/UMPU tests.
- *Comparison.* LRT and score behave similarly. Wald is worst: it divides by $\hat{p}(1-\hat{p})$, which is unstable near 0 or 1 (e.g. $n = 40$, H_{both}: size 0.081).
- *Changing* p_0. The conclusion **does** change. For $p_0 = 0.1$, T is skewed. For $n = 5, 10$ Wald's size for H_{left} is $0.59, 0.35$ (since $\hat{p} = 0$ has probability 0.9^n), while LRT and score cannot reject H_{left} at all (size 0). Even at $n = 160$ the sizes are 0.067/0.067/0.036 (left) and 0.049/0.029/0.049 (right). Near the boundary, the asymptotic tests need a much larger n; the exact UMP/UMPU tests are not affected.

	H_{left}			H_{right}			H_{both}		
n	LRT	Wald	Score	LRT	Wald	Score	LRT	Wald	Score
main choice: $p_0 = 0.5$									
5	0.031	0.188	0.031	0.031	0.188	0.031	0.062	0.062	0.062
10	0.055	0.055	0.055	0.055	0.055	0.055	0.109	0.109	0.021
20	0.058	0.058	0.058	0.058	0.058	0.058	0.041	0.041	0.041
40	0.040	0.040	0.040	0.040	0.040	0.040	0.038	0.081	0.038
80	0.046	0.046	0.046	0.046	0.046	0.046	0.057	0.057	0.057
160	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048
second choice: $p_0 = 0.1$									
5	0.000	0.590	0.000	0.081	0.009	0.081	0.009	0.599	0.081
10	0.000	0.349	0.000	0.070	0.013	0.070	0.013	0.350	0.070
20	0.122	0.122	0.000	0.043	0.011	0.043	0.133	0.124	0.043
40	0.080	0.080	0.015	0.042	0.015	0.042	0.030	0.086	0.057
80	0.035	0.088	0.035	0.054	0.027	0.054	0.062	0.100	0.037
160	0.067	0.067	0.036	0.049	0.029	0.049	0.065	0.045	0.046

Table 1: Exact sizes of the three tests (nominal level 0.05).

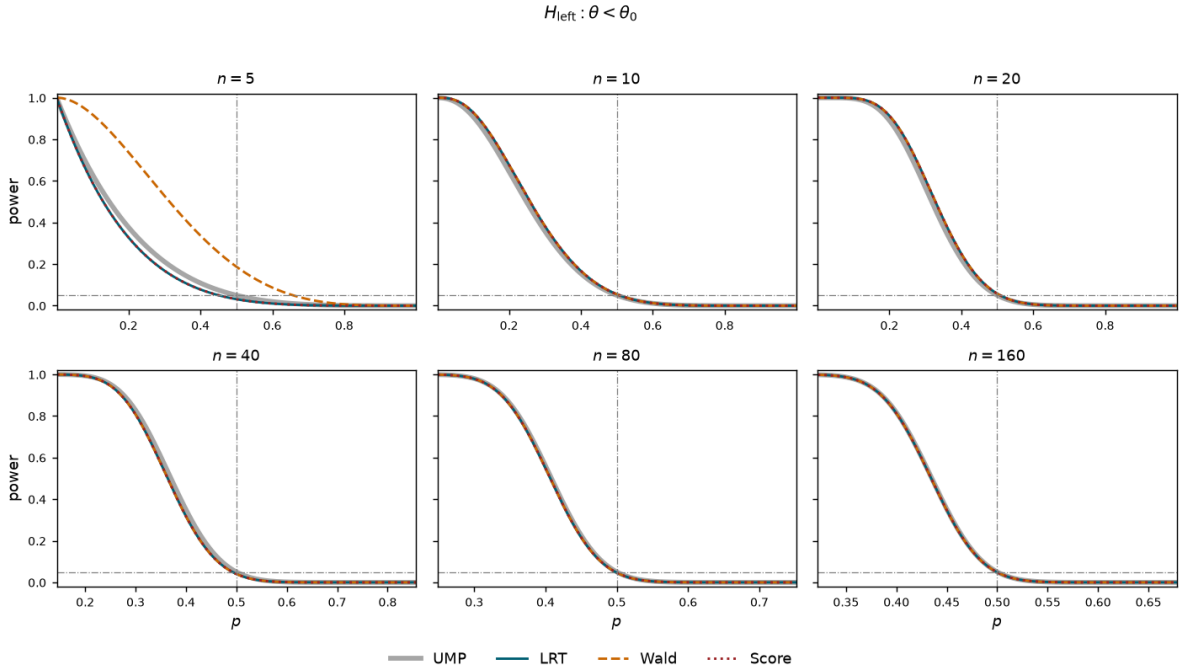


Figure 1: Left alternative ($p_0 = 0.5$).

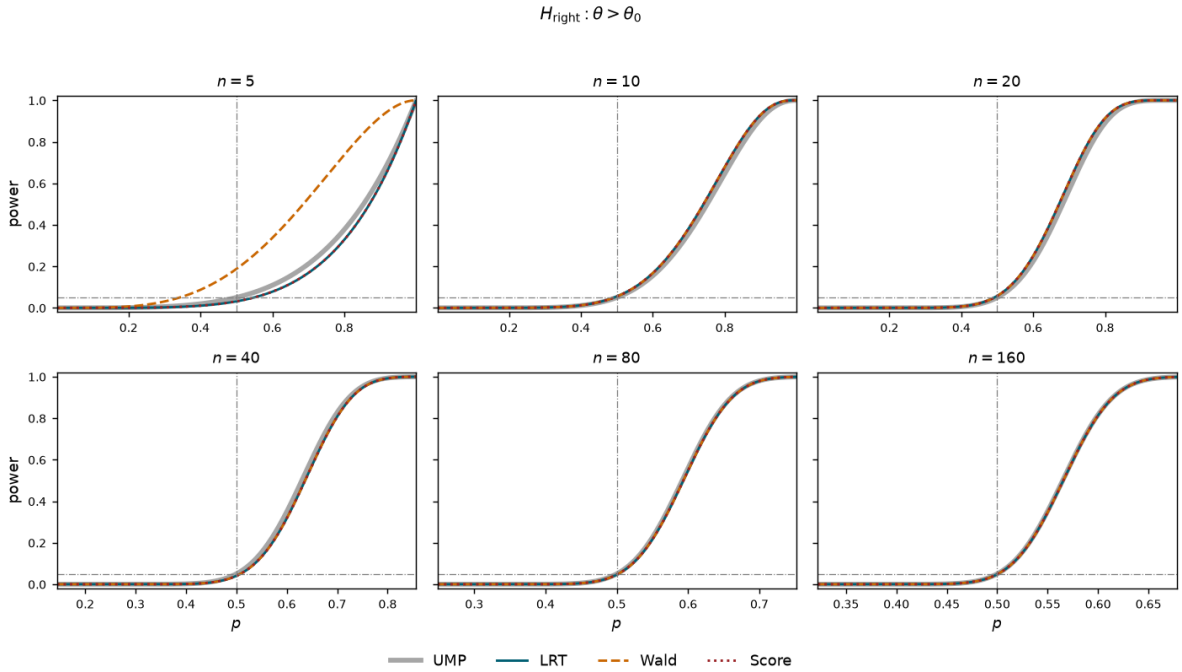


Figure 2: Right alternative ($p_0 = 0.5$).

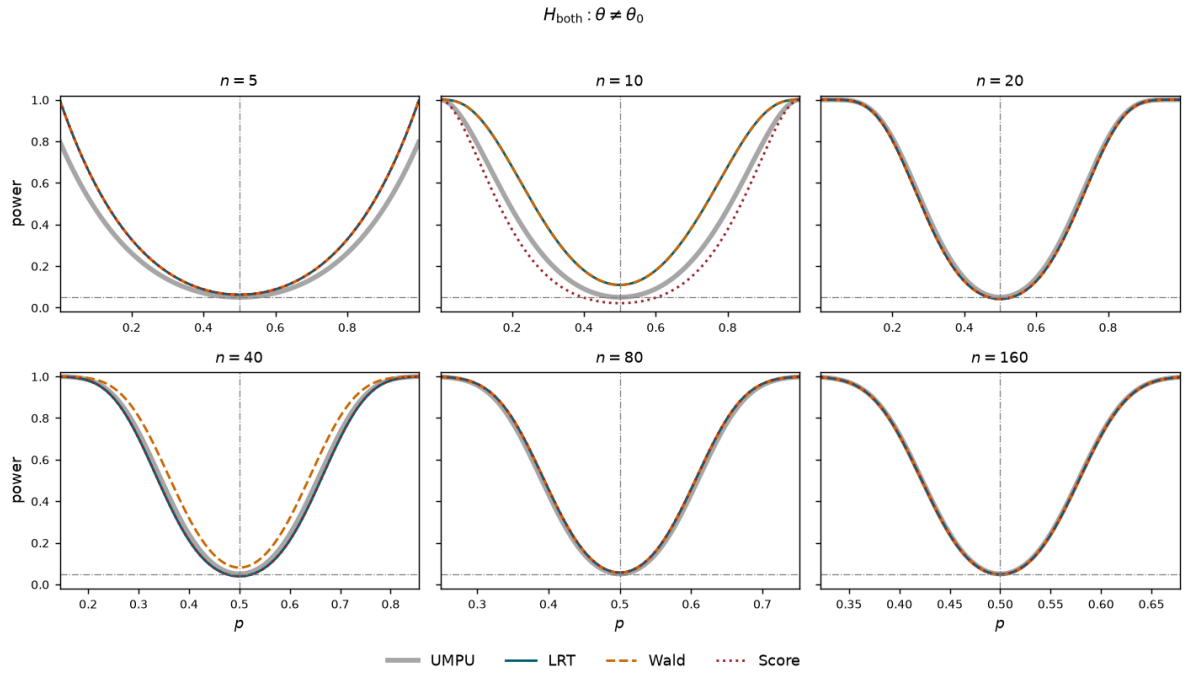


Figure 3: Two-sided alternative ($p_0 = 0.5$).

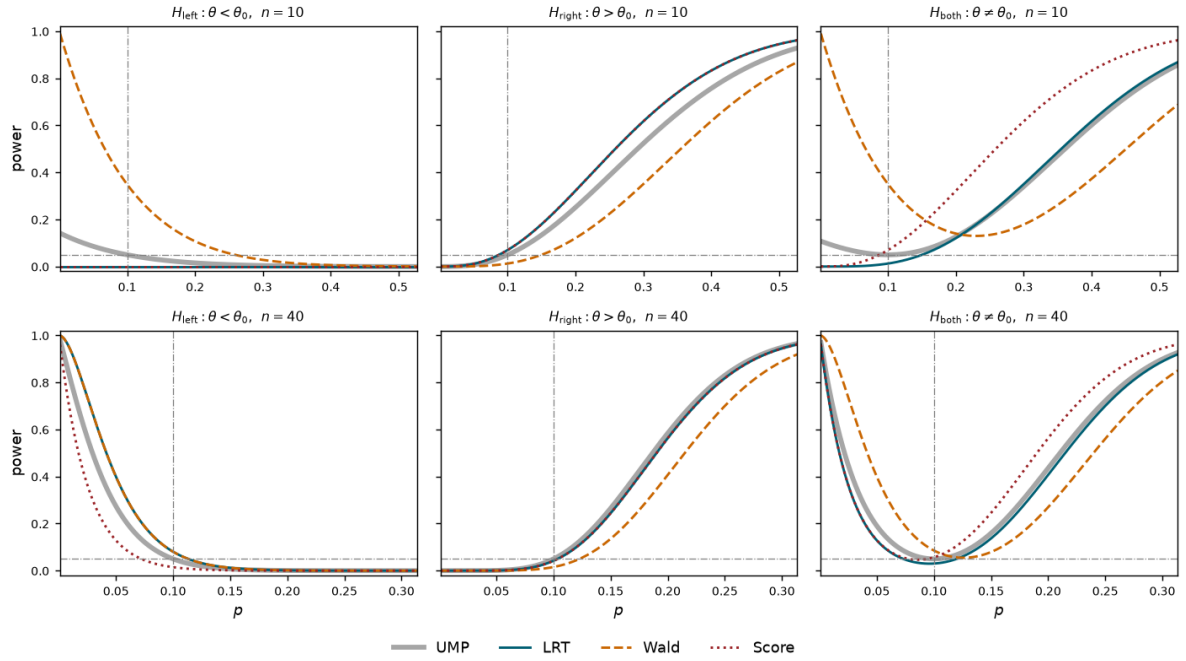


Figure 4: Changed θ_0 ($p_0 = 0.1$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(b) $T = \sum X_i \sim \text{Poisson}(n\lambda)$ is sufficient, and

$$\ell(\lambda) = T \log \lambda - n\lambda + \text{const}, \qquad U_n(\lambda) = \frac{n(\hat{\lambda} - \lambda)}{\lambda}, \qquad I(\lambda) = \frac{1}{\lambda}, \qquad \hat{\lambda} = \bar{X}_n$$

Test statistics:

$$R_n = \operatorname{sgn}(\hat\lambda - \lambda_0)\sqrt{2n\left[\hat\lambda \log\frac{\hat\lambda}{\lambda_0} - (\hat\lambda - \lambda_0)\right]}$$

$$W_n = \frac{\sqrt{n}(\hat\lambda - \lambda_0)}{\sqrt{\hat\lambda}}, \qquad S_n = \frac{\sqrt{n}(\hat\lambda - \lambda_0)}{\sqrt{\lambda_0}}$$

with $W_n = \pm\infty$ if $\hat\lambda = 0$.

Optimal tests. $a(\lambda) = \log\lambda$ is increasing, so there is MLR in T. The UMP tests reject for large T (H_{right}) or small T (H_{left}), randomizing at the boundary so that the size under Poisson($n\lambda_0$) is exactly α. The UMPU test rejects outside $[c_1, c_2]$ with randomization at c_1, c_2 such that $E_{\lambda_0}[\phi] = \alpha$ and $E_{\lambda_0}[T\phi] = \alpha n\lambda_0$.

Observations.

- The score test is the usual Poisson z-test. The pattern is like (a). For $\lambda_0 = 1$, $n = 5$, the H_{both} sizes are 0.072/0.130/0.039 (LRT/Wald/score); at $n = 160$ they are 0.048/0.055/0.053.
- Wald is liberal on the left and conservative on the right: sizes 0.125 and 0.014 at $n = 5$, 0.059 and 0.040 at $n = 160$. The reason is $\sqrt{\hat\lambda}$ in the denominator: it is small when $\hat\lambda < \lambda_0$ (so W_n is inflated) and large when $\hat\lambda > \lambda_0$.
- The LRT is closest to the optimal tests, then score. Wald's maximum shortfall from the UMPU test is 0.15 at $n = 10$ and still 0.03 at $n = 160$.
- *Changing* λ_0. The conclusion **does** change. For $\lambda_0 = 0.2$, $n\lambda_0$ is small (1 at $n = 5$). LRT and score have size 0 for H_{left} at $n = 5$ (score also at $n = 10$), and Wald has size 0.368 (as $\hat\lambda = 0$ has probability e^{-1}). The smaller $n\lambda_0$ is, the worse the asymptotic tests are.

	H_{left}			H_{right}			H_{both}		
n	LRT	Wald	Score	LRT	Wald	Score	LRT	Wald	Score
main choice: $\lambda_0 = 1$									
5	0.040	0.125	0.040	0.032	0.014	0.068	0.072	0.130	0.039
10	0.067	0.067	0.029	0.049	0.027	0.049	0.056	0.074	0.037
20	0.066	0.066	0.039	0.052	0.034	0.052	0.043	0.052	0.056
40	0.062	0.062	0.043	0.053	0.039	0.053	0.049	0.057	0.047
80	0.049	0.062	0.049	0.045	0.036	0.056	0.051	0.055	0.050
160	0.050	0.059	0.050	0.047	0.040	0.055	0.048	0.055	0.053
second choice: $\lambda_0 = 0.2$									
5	0.000	0.368	0.000	0.019	0.004	0.080	0.019	0.368	0.080
10	0.135	0.135	0.000	0.053	0.005	0.053	0.152	0.136	0.053
20	0.092	0.092	0.018	0.051	0.021	0.051	0.040	0.094	0.069
40	0.042	0.100	0.042	0.034	0.017	0.064	0.060	0.108	0.048
80	0.043	0.077	0.043	0.037	0.022	0.058	0.044	0.056	0.059
160	0.061	0.061	0.041	0.051	0.036	0.051	0.051	0.052	0.041

Table 2: Exact sizes of the three tests (nominal level 0.05).

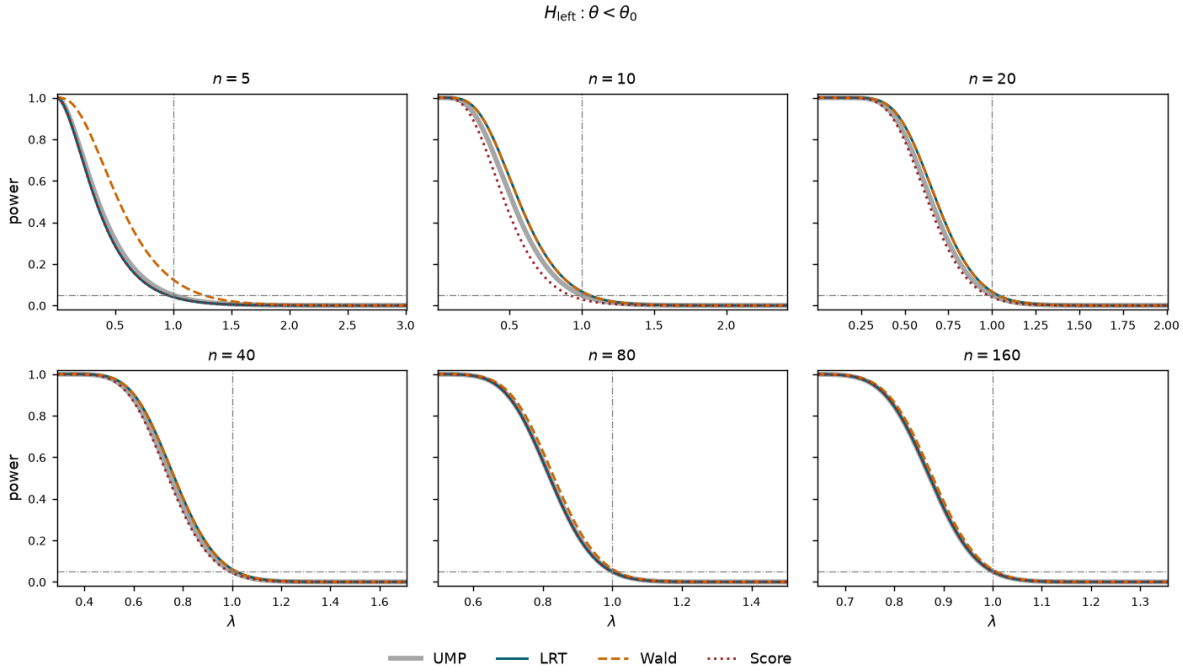


Figure 5: Left alternative ($\lambda_0 = 1$).

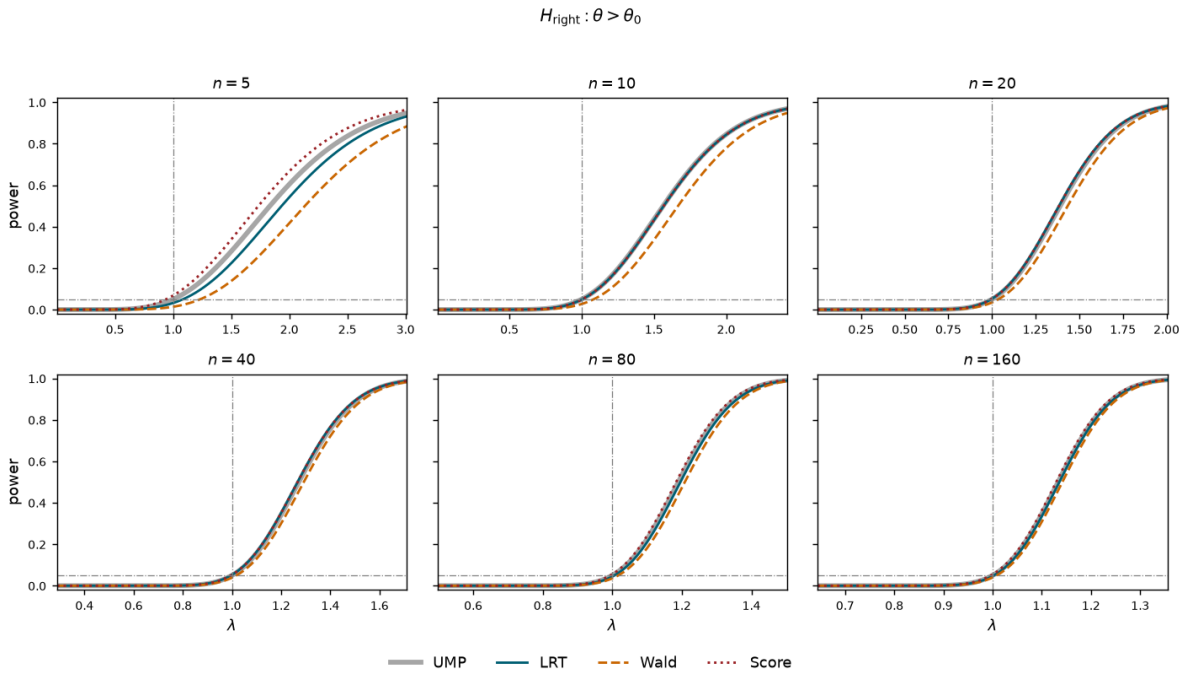


Figure 6: Right alternative ($\lambda_0 = 1$).

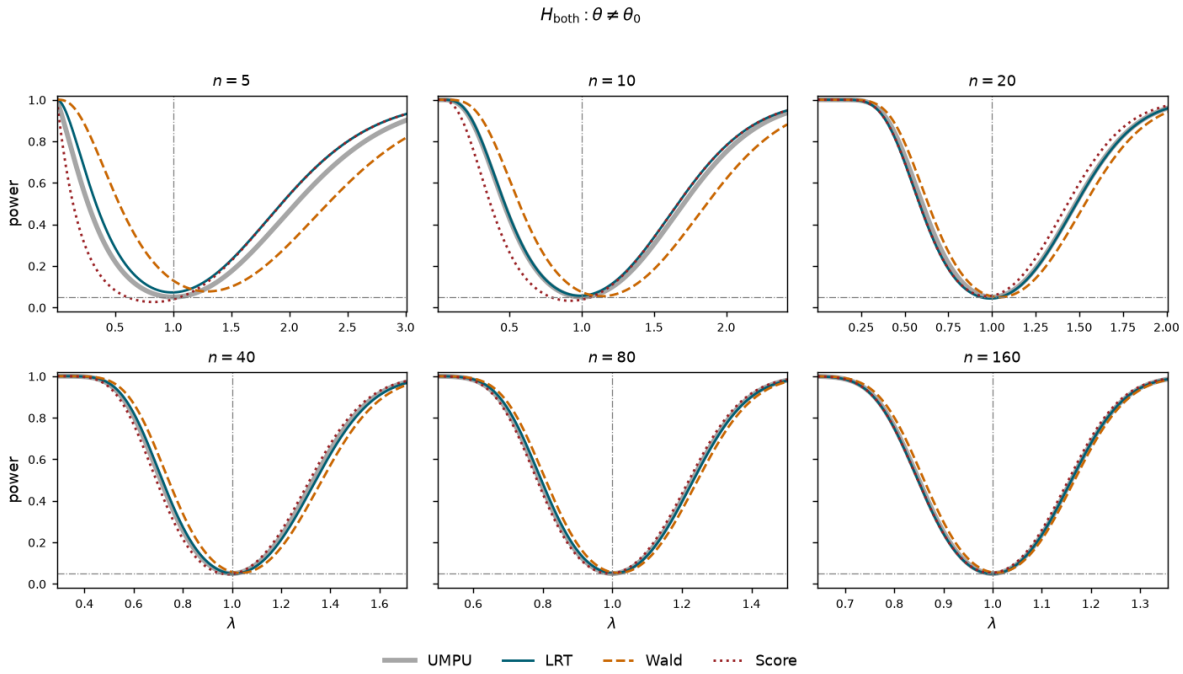


Figure 7: Two-sided alternative ($\lambda_0 = 1$).

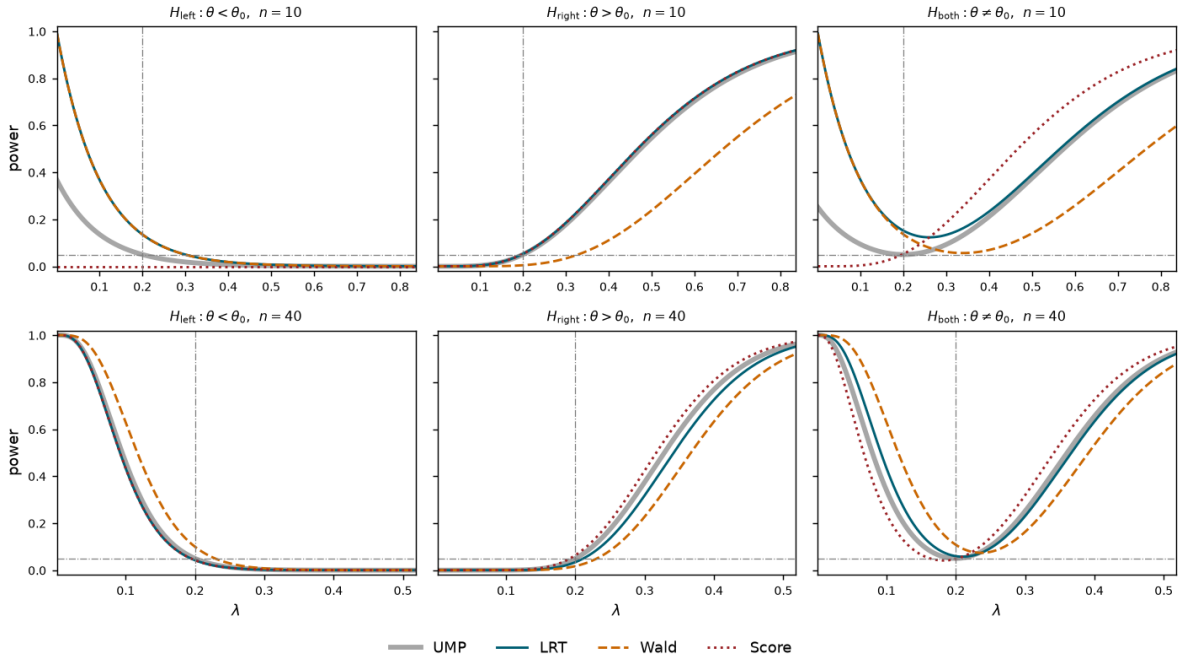


Figure 8: Changed θ_0 ($\lambda_0 = 0.2$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(c) The pmf is $f_p(x) = p(1-p)^{x-1}$, $x = 1, 2, \ldots$. $T = \sum X_i \geq n$ is sufficient, with $T - n \sim \text{NegBin}(n, p)$, and

$$\ell(p) = n \log p + (T - n) \log(1 - p), \qquad U_n(p) = \frac{n - pT}{p(1-p)}, \qquad I(p) = \frac{1}{p^2(1-p)}, \qquad \hat{p} = \frac{n}{T} = \frac{1}{\bar{X}_n}$$

Test statistics:

$$R_n = \operatorname{sgn}(\hat{p} - p_0)\sqrt{2n\left[\log\frac{\hat{p}}{p_0} + \left(\frac{1}{\hat{p}} - 1\right)\log\frac{1-\hat{p}}{1-p_0}\right]}$$

$$W_n = \frac{\sqrt{n}(\hat{p} - p_0)}{\hat{p}\sqrt{1-\hat{p}}}, \qquad S_n = \frac{\sqrt{n}\,(1 - p_0\bar{X}_n)}{\sqrt{1-p_0}} = \frac{\sqrt{n}(\hat{p} - p_0)}{\hat{p}\sqrt{1-p_0}}$$

with $W_n = \pm\infty$ if $\hat{p} = 1$.

Optimal tests. The pmf is $\propto \exp\{\log(1-p)(T-n)\}$ and $a(p) = \log(1-p)$ is *decreasing*, so there is MLR in $-T$. The UMP test for H_{right} rejects for *small* T, and the UMP test for H_{left} rejects for *large* T (both randomized at the boundary). The UMPU test rejects outside $[c_1, c_2]$ with randomization at c_1, c_2 such that $E_{p_0}[\phi] = \alpha$ and $E_{p_0}[T\phi] = \alpha n/p_0$.

Observations.

- For $p_0 = 0.5$ the three tests are close from $n = 20$ on, and sizes are within $[0.041, 0.061]$ for $n \geq 40$.
- The score test is the most erratic at small n. For H_{right} and $n = 5$ we always have $S_n \leq \sqrt{n(1-p_0)} = 1.58 < 1.645$, so it *never rejects* (size and power 0), while LRT and Wald have size 0.109. For H_{both}, Wald has size 0.134 at $n = 5$ and the LRT 0.047.
- The LRT is closest to the UMP/UMPU tests (shortfall ≤ 0.03 for $n \geq 10$). The score test loses up to 0.33 to the UMPU test at $n = 10$, partly because its size is below α there.
- *Changing* p_0. The conclusion **does** change. For $p_0 = 0.9$, $\hat{p} = 1$ has probability 0.9^n. Wald has size 0.59 ($n = 5$) and 0.35 ($n = 10$) for H_{right}, since $W_n = +\infty$ there, while LRT and score have size 0. Even at $n = 160$ the H_{right} sizes are 0.071/0.071/0.041.

	H_{left}			H_{right}			H_{both}		
n	LRT	Wald	Score	LRT	Wald	Score	LRT	Wald	Score
main choice: $p_0 = 0.5$									
5	0.038	0.038	0.059	0.109	0.109	0.000	0.047	0.134	0.038
10	0.044	0.044	0.061	0.046	0.090	0.019	0.068	0.068	0.049
20	0.046	0.046	0.059	0.049	0.075	0.031	0.051	0.057	0.045
40	0.041	0.050	0.061	0.054	0.054	0.041	0.052	0.052	0.049
80	0.044	0.044	0.059	0.054	0.054	0.045	0.054	0.054	0.052
160	0.049	0.049	0.054	0.050	0.057	0.044	0.050	0.054	0.047
second choice: $p_0 = 0.9$									
5	0.026	0.005	0.114	0.000	0.590	0.000	0.026	0.596	0.026
10	0.034	0.009	0.111	0.000	0.349	0.000	0.009	0.351	0.034
20	0.033	0.012	0.085	0.122	0.122	0.000	0.133	0.125	0.033
40	0.046	0.022	0.046	0.074	0.074	0.015	0.036	0.083	0.061
80	0.046	0.026	0.046	0.069	0.069	0.028	0.055	0.076	0.035
160	0.048	0.032	0.048	0.071	0.071	0.041	0.042	0.054	0.054

Table 3: Exact sizes of the three tests (nominal level 0.05).

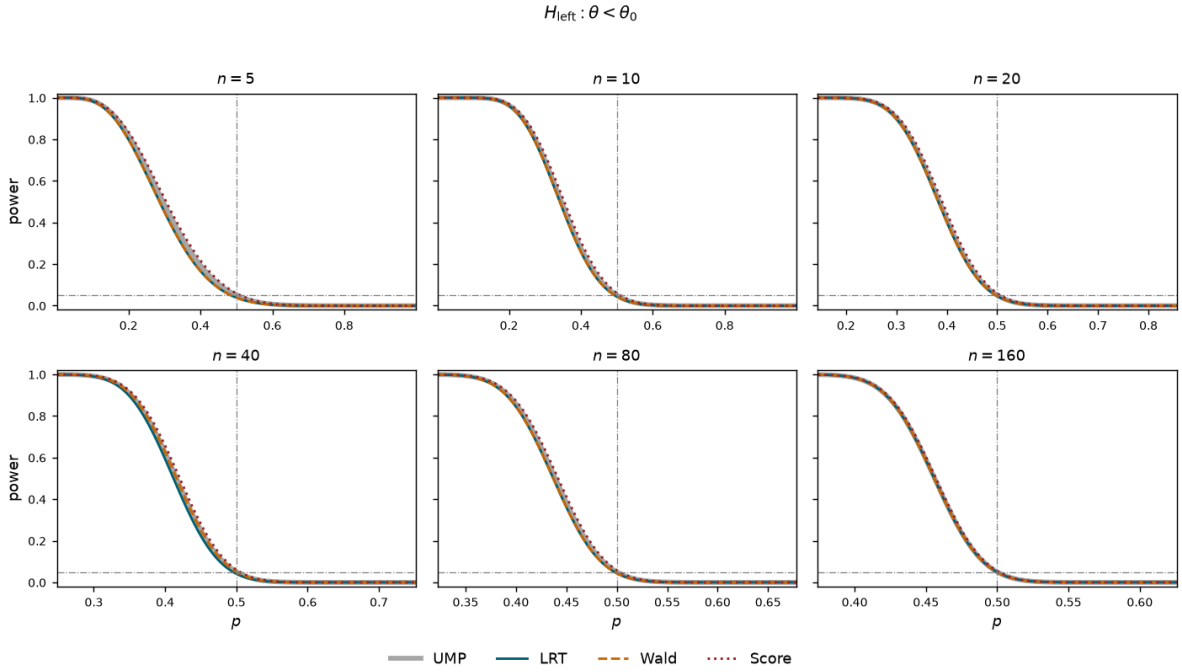


Figure 9: Left alternative ($p_0 = 0.5$).

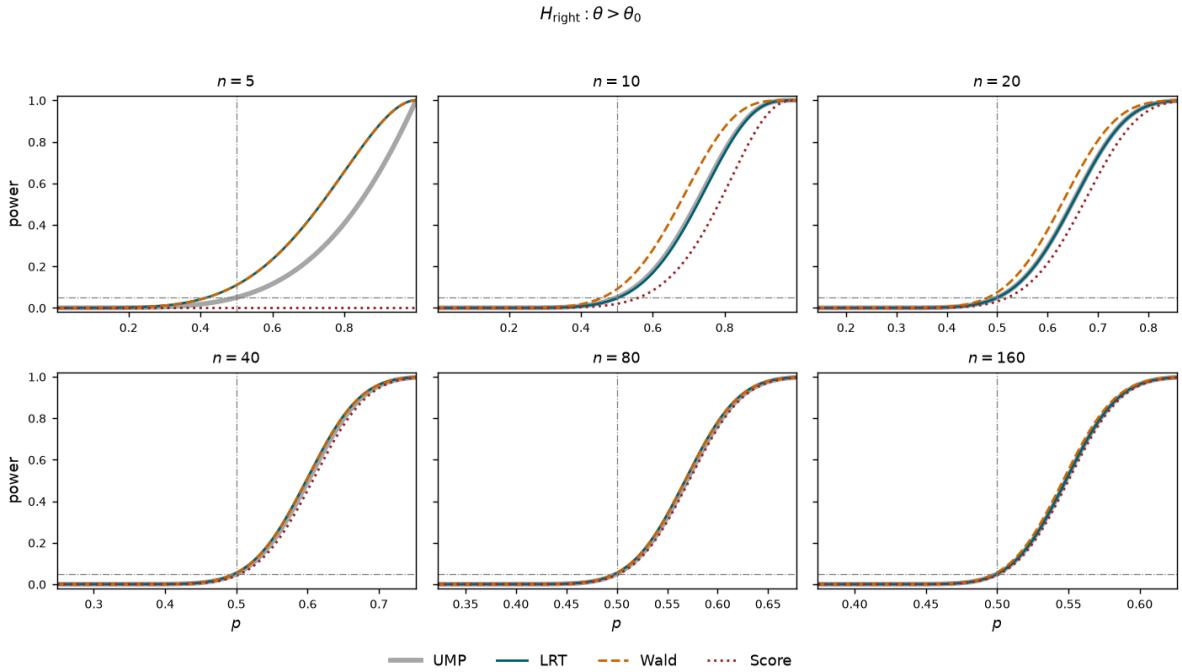


Figure 10: Right alternative ($p_0 = 0.5$).

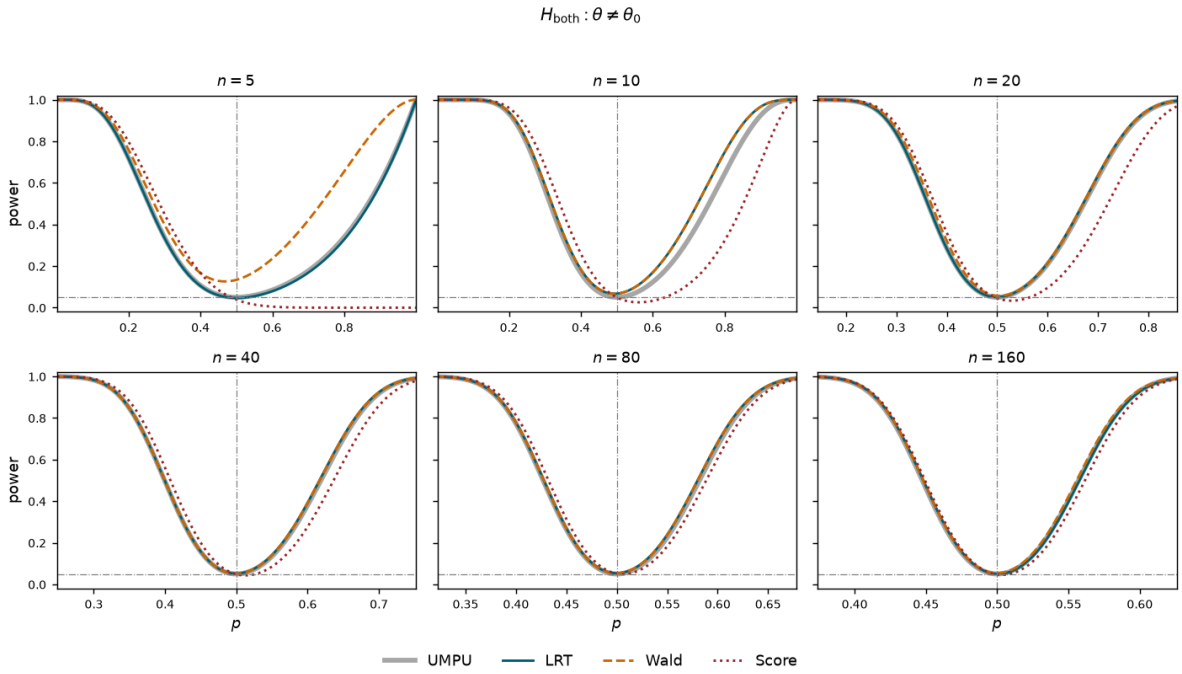


Figure 11: Two-sided alternative ($p_0 = 0.5$).

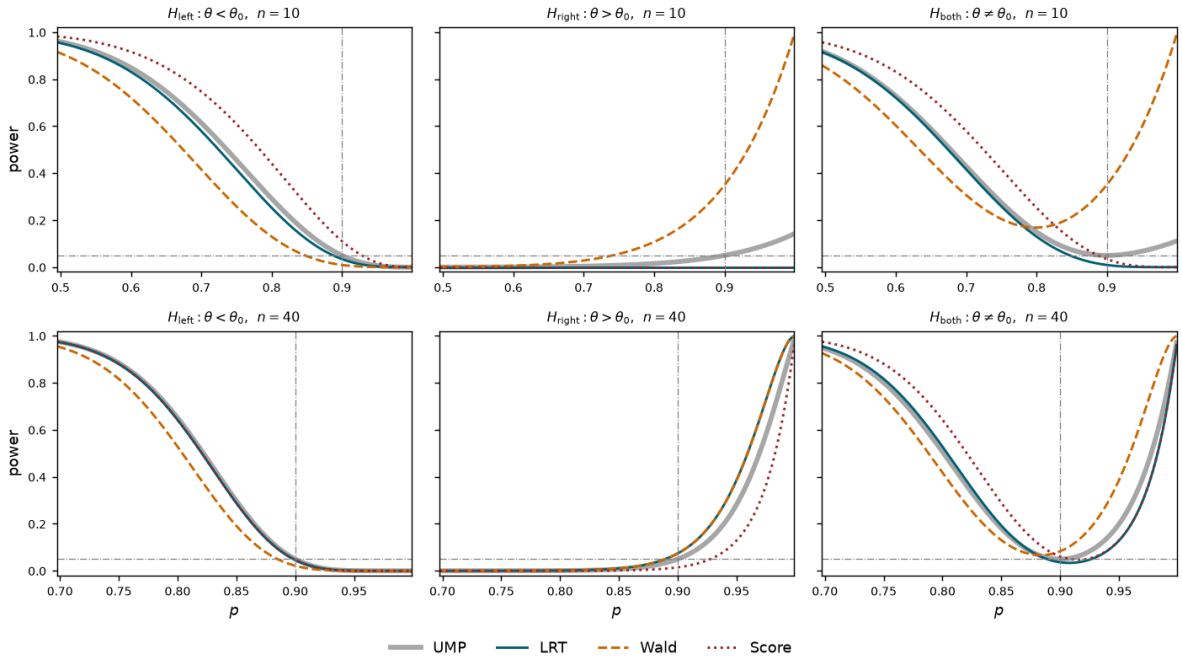


Figure 12: Changed θ_0 ($p_0 = 0.9$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(d) This model is not regular: the support depends on θ. The likelihood is $L(\theta) = \theta^{-n}\mathbb{1}_{\{X_{(n)} \leq \theta\}}$, which is 0 for $\theta < X_{(n)}$ and decreasing after that. So $\hat{\theta} = X_{(n)}$, with $\mathbb{P}_\theta(X_{(n)} \leq x) = (x/\theta)^n$ on $[0, \theta]$. Under H_0, $(X_{(n)}/\theta_0)^n \sim U(0, 1)$.

LRT. If $X_{(n)} > \theta_0$ then $L(\theta_0) = 0$, so we reject. If $X_{(n)} \leq \theta_0$ then $\Lambda_n = (X_{(n)}/\theta_0)^n \sim U(0, 1)$ under H_0. Hence

$$-2 \log \Lambda_n \sim \chi^2_2 \quad \text{exactly, for every } n,$$

not χ^2_1. Using the cut-off 3.84 would give size $P(\chi^2_2 > 3.84) = e^{-1.92} = 0.147$, so we use the exact cut-off $\Lambda_n \le \alpha$ instead:

- H_{both}: reject if $X_{(n)} > \theta_0$ or $X_{(n)} \le \alpha^{1/n}\theta_0$.
- H_{left}: reject if $X_{(n)} \le \alpha^{1/n}\theta_0$.
- H_{right}: the restricted MLE is $\max(X_{(n)}, \theta_0)$, so $\Lambda_n = 1$ if $X_{(n)} \le \theta_0$ and 0 otherwise. The LRT rejects iff $X_{(n)} > \theta_0$, which has size 0 (the cut-off cannot be tuned, as Λ_n takes only two values).

Wald and score. The score $\partial_\theta \log f_\theta(x) = -1/\theta$ is not random, so $U_n(\theta_0) = -n/\theta_0$ and $E[U] = 0$ fails. Using the formal information $I(\theta) = 1/\theta^2$,

$$W_n = \frac{\sqrt{n}(X_{(n)} - \theta_0)}{X_{(n)}}, \qquad S_n = \frac{-n/\theta_0}{\sqrt{n}/\theta_0} = -\sqrt{n}$$

The score statistic is a *constant*, so it carries no information. For $n \ge 4$ the score test for H_{left} and H_{both} **always rejects** (size 1). For H_{right} it rejects only when $X_{(n)} > \theta_0$. Wald is a real test: it rejects H_{left} iff $X_{(n)} < \theta_0/(1 + z_{0.95}/\sqrt{n})$, and H_{right} iff $X_{(n)} > \theta_0/(1 - z_{0.95}/\sqrt{n})$ (impossible if $\sqrt{n} \le z_{0.95}$). For H_{both} it uses $z_{0.975}$ on both sides.

Optimal tests. There is MLR in $X_{(n)}$. For H_{right} the UMP test rejects if $X_{(n)} > \theta_0(1-\alpha)^{1/n}$, with power $1-(1-\alpha)(\theta_0/\theta)^n$ for $\theta > \theta_0$. For H_{left} the UMP test rejects if $X_{(n)} < \theta_0\alpha^{1/n}$. For H_{both} the UMPU test rejects if $X_{(n)} > \theta_0$ or $X_{(n)} < \theta_0\alpha^{1/n}$.

Observations.

- The exact LRT *equals* the UMP test for H_{left} and the UMPU test for H_{both} (blue and grey curves coincide). For H_{right} it has size 0 and loses at most $\alpha = 0.05$ against the UMP test (its power is $1 - (\theta_0/\theta)^n$).
- Wald is bad: its size goes to 0 (for H_{both}: 0.043 at $n = 5$, at most 0.008 for $n \ge 10$). The reason is that $X_{(n)}$ converges at rate $1/n$, not $1/\sqrt{n}$. Wald only detects alternatives at distance about $\theta_0 z/\sqrt{n}$, while the UMPU test detects them at distance about θ_0/n.
- The score test is useless (constant statistic). The classical tests rely on regularity, which fails here; the LRT with the exact cut-off is optimal.
- This is a scale family, so every power function depends only on θ/θ_0. The conclusion does **not** change with θ_0.

n	H_{left} LRT	Wald	Score	H_{right} LRT	Wald	Score	H_{both} LRT	Wald	Score
main choice: $\theta_0 = 1$									
5	0.050	0.063	1.000	0.000	0.000	0.000	0.050	0.043	1.000
10	0.050	0.015	1.000	0.000	0.000	0.000	0.050	0.008	1.000
20	0.050	0.002	1.000	0.000	0.000	0.000	0.050	0.001	1.000
40	0.050	0.000	1.000	0.000	0.000	0.000	0.050	0.000	1.000
80	0.050	0.000	1.000	0.000	0.000	0.000	0.050	0.000	1.000
160	0.050	0.000	1.000	0.000	0.000	0.000	0.050	0.000	1.000
second choice: $\theta_0 = 5$									
5	0.050	0.063	1.000	0.000	0.000	0.000	0.050	0.043	1.000
10	0.050	0.015	1.000	0.000	0.000	0.000	0.050	0.008	1.000
20	0.050	0.002	1.000	0.000	0.000	0.000	0.050	0.001	1.000
40	0.050	0.000	1.000	0.000	0.000	0.000	0.050	0.000	1.000
80	0.050	0.000	1.000	0.000	0.000	0.000	0.050	0.000	1.000
160	0.050	0.000	1.000	0.000	0.000	0.000	0.050	0.000	1.000

Table 4: Exact sizes of the three tests (nominal level 0.05).

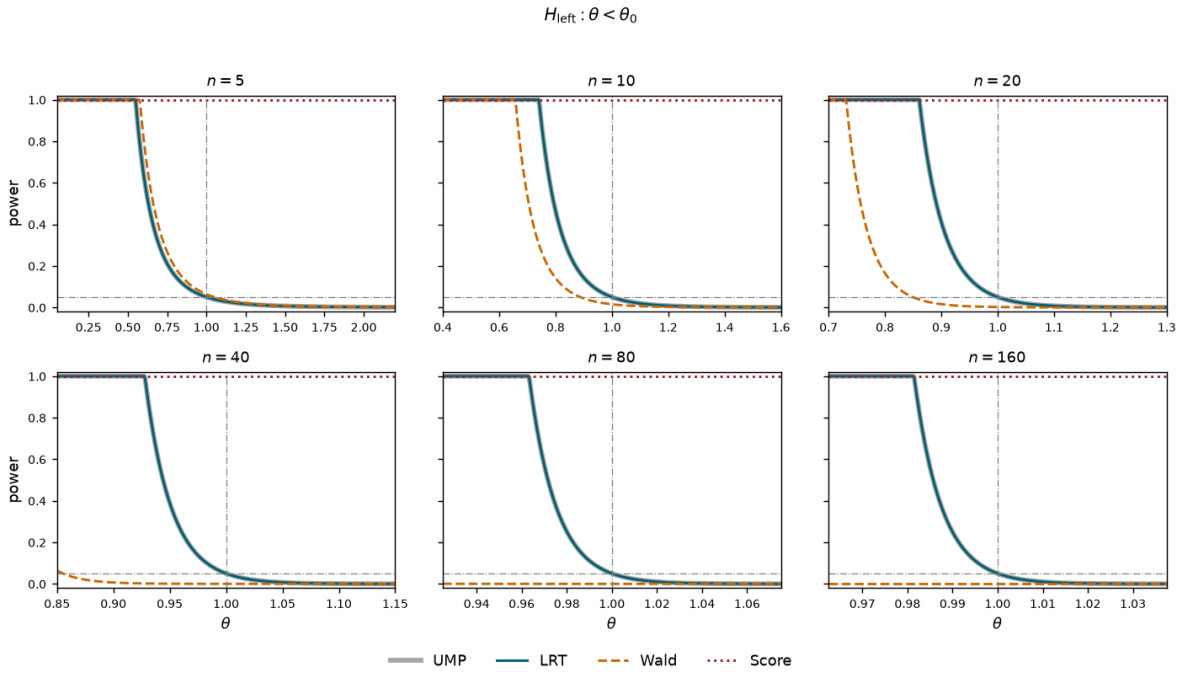


Figure 13: Left alternative ($\theta_0 = 1$).

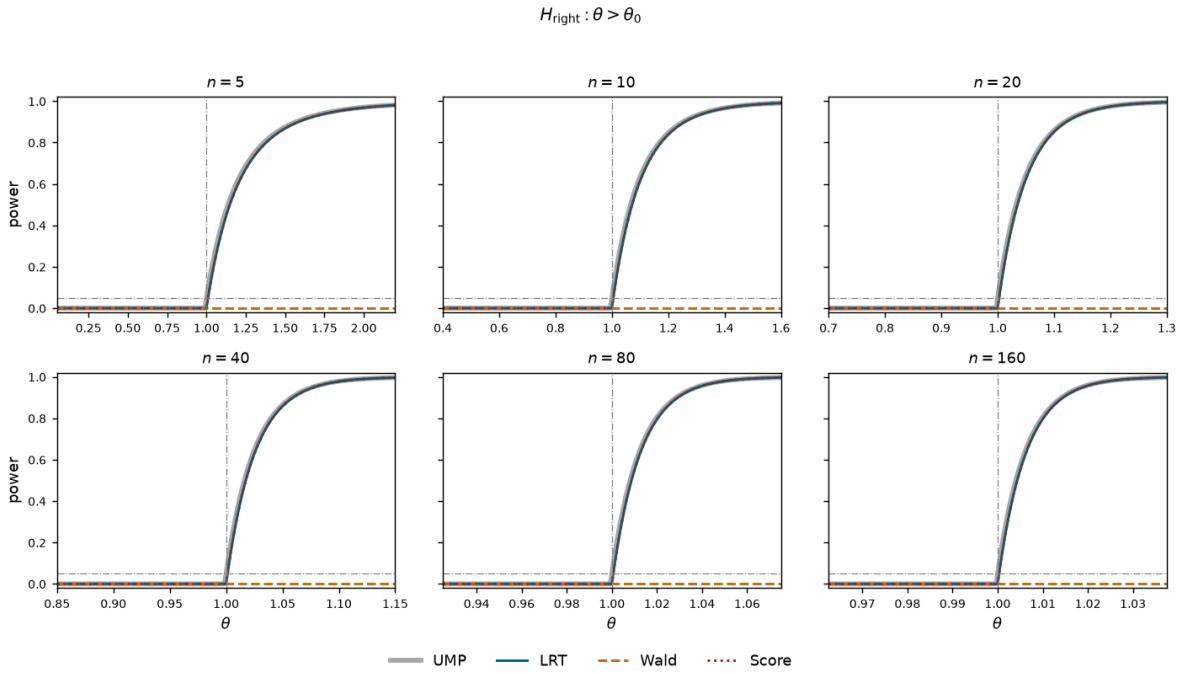


Figure 14: Right alternative ($\theta_0 = 1$).

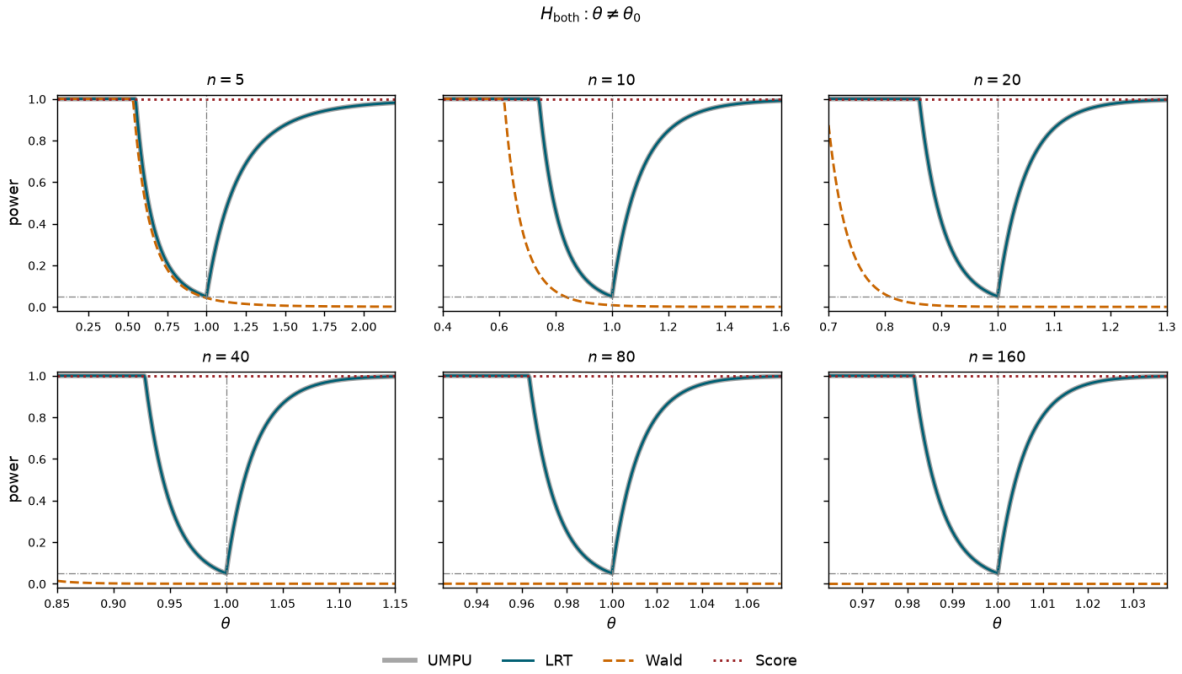


Figure 15: Two-sided alternative ($\theta_0 = 1$).

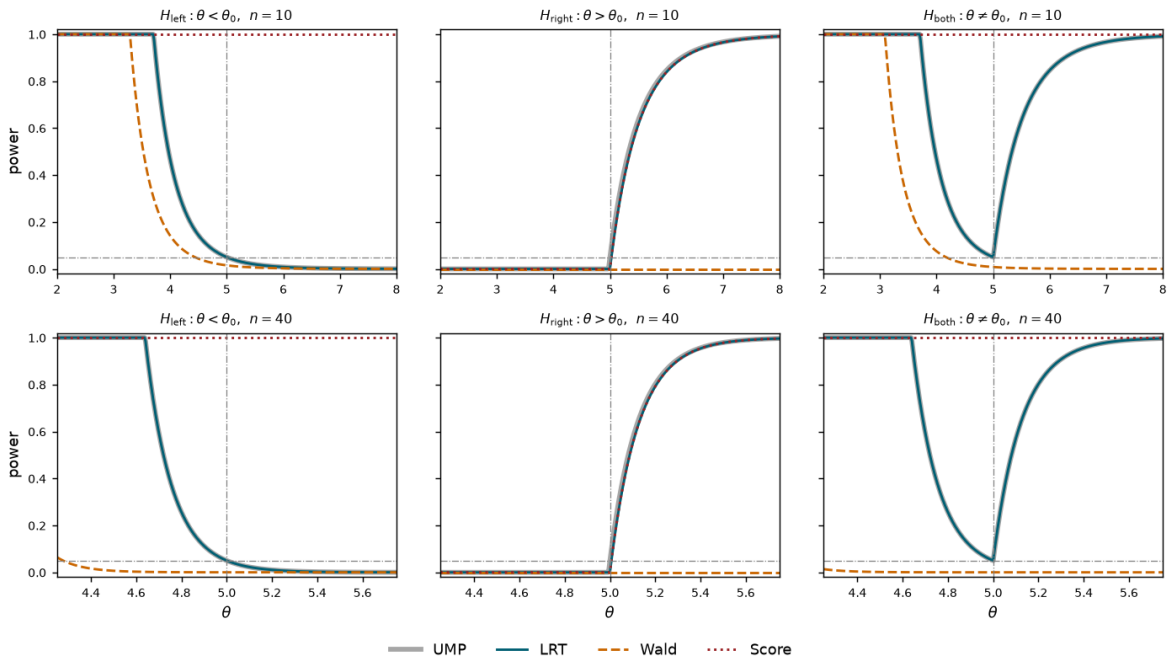


Figure 16: Changed θ_0 ($\theta_0 = 5$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(e) Here

$$\ell(\mu) = -\frac{1}{2}\sum (x_i - \mu)^2 + \text{const}, \qquad U_n(\mu) = n(\bar{X}_n - \mu), \qquad I(\mu) = 1, \qquad \hat{\mu} = \bar{X}_n$$

so $-2\log\Lambda_n = n(\bar{X}_n - \mu_0)^2$ and

$$R_n = W_n = S_n = Z := \sqrt{n}(\bar{X}_n - \mu_0) \sim N(0,1) \text{ under } H_0$$

All three tests are **identical**, and Z is exactly $N(0,1)$ for every n. The tests are: reject H_{right} if $Z > z_{0.95}$, H_{left} if $Z < -z_{0.95}$, H_{both} if $|Z| > z_{0.975}$. The power functions are

$$\beta_{\text{right}}(\mu) = 1 - \Phi\big(z_{0.95} - \sqrt{n}(\mu - \mu_0)\big), \qquad \beta_{\text{left}}(\mu) = \Phi\big(-z_{0.95} - \sqrt{n}(\mu - \mu_0)\big)$$
$$\beta_{\text{both}}(\mu) = 1 - \Phi\big(z_{0.975} - \sqrt{n}(\mu - \mu_0)\big) + \Phi\big(-z_{0.975} - \sqrt{n}(\mu - \mu_0)\big)$$

Optimal tests. The density is $\propto \exp\{\mu \sum x_i - n\mu^2/2\}$ with $a(\mu) = \mu$ increasing, so there is MLR in $\sum X_i$. The UMP tests are the one-sided z-tests above. The UMPU test is the two-sided z-test: Z is symmetric about 0 under H_0, so symmetric cut-offs satisfy $E_{\mu_0}[\phi] = \alpha$ and $E_{\mu_0}[T\phi] = \alpha E_{\mu_0}[T]$.

Observations. All four curves coincide in every figure, and all sizes equal 0.05 exactly. There is nothing to choose between the tests. A larger n only makes the curves steeper (the horizontal scale is $1/\sqrt{n}$). This is a location family, so $\beta(\mu)$ depends only on $\mu - \mu_0$, and the conclusion does **not** change with μ_0.

n	H_{left} LRT	Wald	Score	H_{right} LRT	Wald	Score	H_{both} LRT	Wald	Score
main choice: $\mu_0 = 0$ *(second choice* $\mu_0 = 3$ *gives the same table)*									
5	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
10	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
20	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
40	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
80	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
160	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050

Table 5: Exact sizes of the three tests (nominal level 0.05).

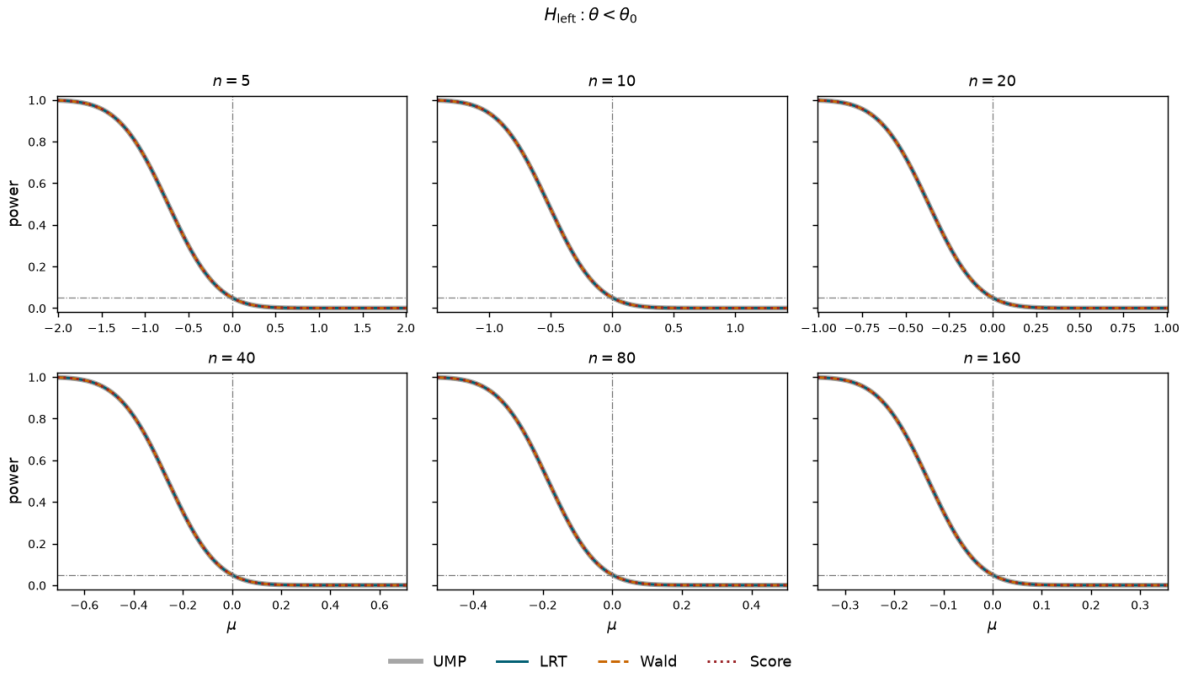


Figure 17: Left alternative ($\mu_0 = 0$).

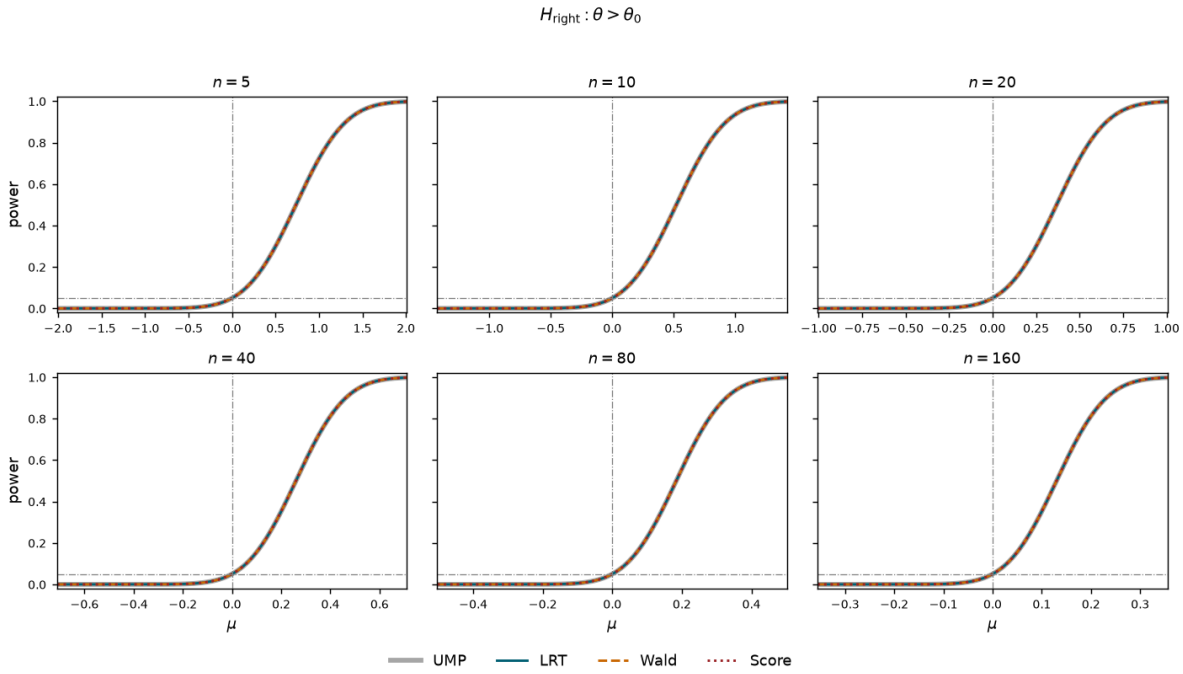


Figure 18: Right alternative ($\mu_0 = 0$).

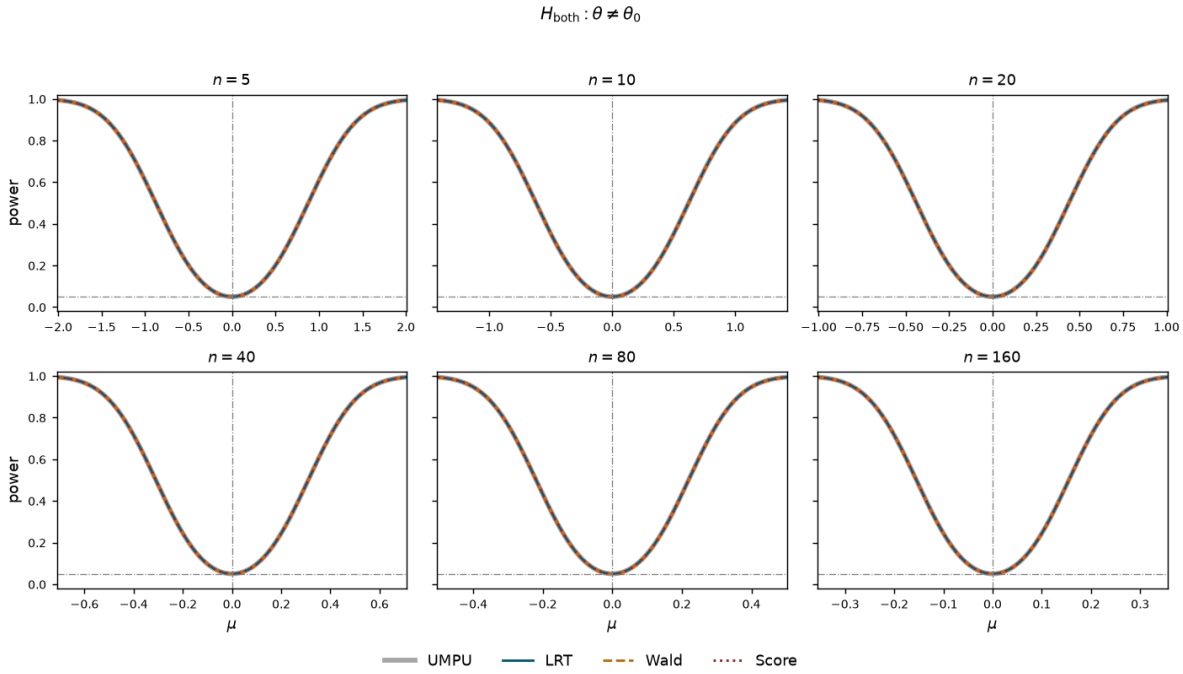


Figure 19: Two-sided alternative ($\mu_0 = 0$).

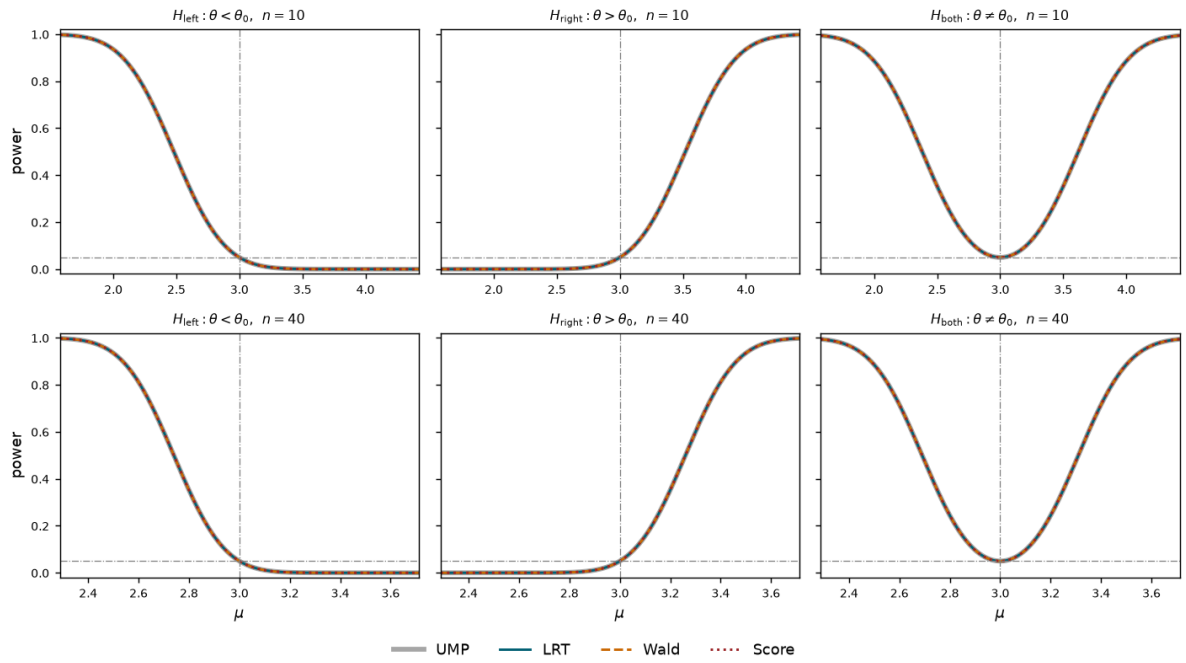


Figure 20: Changed θ_0 ($\mu_0 = 3$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(f) Let $\theta = \sigma^2$ and $T = \sum X_i^2$. Then $\hat\theta = T/n$ and $T/\theta \sim \chi^2_n$, and

$$\ell(\theta) = -\frac{n}{2}\log\theta - \frac{T}{2\theta} + \text{const}, \qquad U_n(\theta) = \frac{n(\hat\theta - \theta)}{2\theta^2}, \qquad I(\theta) = \frac{1}{2\theta^2}$$

Test statistics:

$$R_n = \operatorname{sgn}(\hat\theta - \theta_0)\sqrt{n\left[\frac{\hat\theta}{\theta_0} - 1 - \log\frac{\hat\theta}{\theta_0}\right]}$$

$$W_n = \sqrt{\frac{n}{2}}\left(1 - \frac{\theta_0}{\hat\theta}\right), \qquad S_n = \sqrt{\frac{n}{2}}\left(\frac{\hat\theta}{\theta_0} - 1\right)$$

Optimal tests. The density is $\propto \theta^{-n/2}\exp\{-T/(2\theta)\}$ with $a(\theta) = -1/(2\theta)$ increasing, so there is MLR in T. The UMP test rejects H_{right} if $T > \theta_0\chi^2_{n,1-\alpha}$ and H_{left} if $T < \theta_0\chi^2_{n,\alpha}$. The UMPU test rejects if $T < \theta_0 w_1$ or $T > \theta_0 w_2$, where $w_1 < w_2$ solve

$$\int_{w_1}^{w_2} f_{\chi^2_n}(w)\,dw = 1 - \alpha, \qquad \int_{w_1}^{w_2} f_{\chi^2_{n+2}}(w)\,dw = 1 - \alpha$$

(the second equation is $E_{\theta_0}[T\phi] = \alpha E_{\theta_0}[T]$, using $w f_{\chi^2_n}(w) = n f_{\chi^2_{n+2}}(w)$). These are *not* the equal-tail quantiles.

Observations.

- The LRT is best and is close to the UMP/UMPU tests for $n \geq 20$ (shortfall ≤ 0.04 for $n \geq 10$). Its sizes (left/right/both) are 0.078/0.033/0.058 at $n = 5$ and 0.054/0.046/0.050 at $n = 160$.

- Wald and score are biased in opposite directions because χ^2_n is skewed. Wald is liberal on the left (size 0.216 at $n = 5$) and very conservative on the right: at $n = 5$, $W_n < \sqrt{n/2} = 1.58 < 1.645$, so it *can never reject* H_{right}. Score is the mirror image: at $n = 5$, $S_n \geq -1.58$, so it can never reject H_{left}, and it is liberal on the right (size 0.07).
- For H_{both} and $n = 10$ the shortfall from the UMPU test is up to 0.70 (Wald) and 0.75 (score), on opposite sides. At $n = 160$ Wald's sizes are still 0.076 (left) and 0.028 (right), while the LRT is within 0.01 of α.
- $\hat{\theta}$ is skewed, and W_n, S_n are simple functions of it, while the LRT adjusts for the skewness. This is a scale family, so the conclusion does **not** change with θ_0.

	H_{left}			H_{right}			H_{both}		
n	LRT	Wald	Score	LRT	Wald	Score	LRT	Wald	Score
main choice: $\sigma_0^2 = 1$ *(second choice* $\sigma_0^2 = 4$ *gives the same table)*									
5	0.078	0.216	0.000	0.033	0.000	0.070	0.058	0.184	0.048
10	0.069	0.165	0.011	0.037	0.000	0.067	0.054	0.132	0.044
20	0.062	0.129	0.025	0.040	0.003	0.064	0.052	0.096	0.045
40	0.058	0.105	0.034	0.043	0.011	0.061	0.051	0.074	0.047
80	0.056	0.088	0.039	0.045	0.020	0.058	0.050	0.062	0.049
160	0.054	0.076	0.043	0.046	0.028	0.056	0.050	0.056	0.049

Table 6: Exact sizes of the three tests (nominal level 0.05).

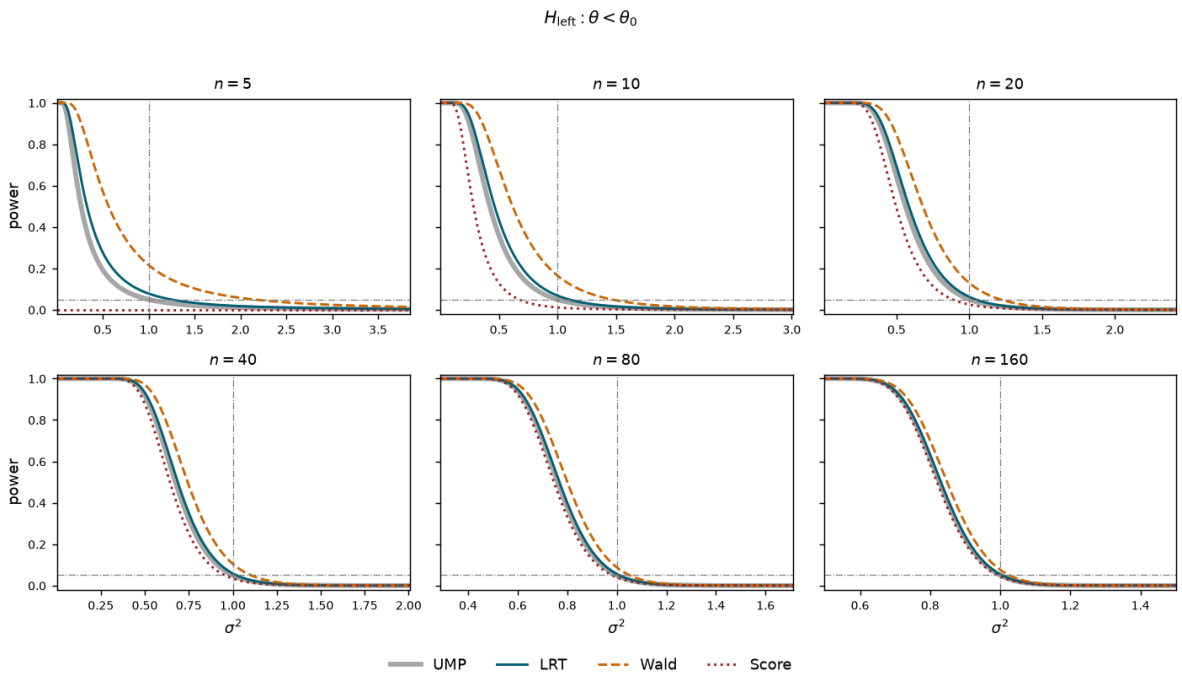


Figure 21: Left alternative ($\sigma_0^2 = 1$).

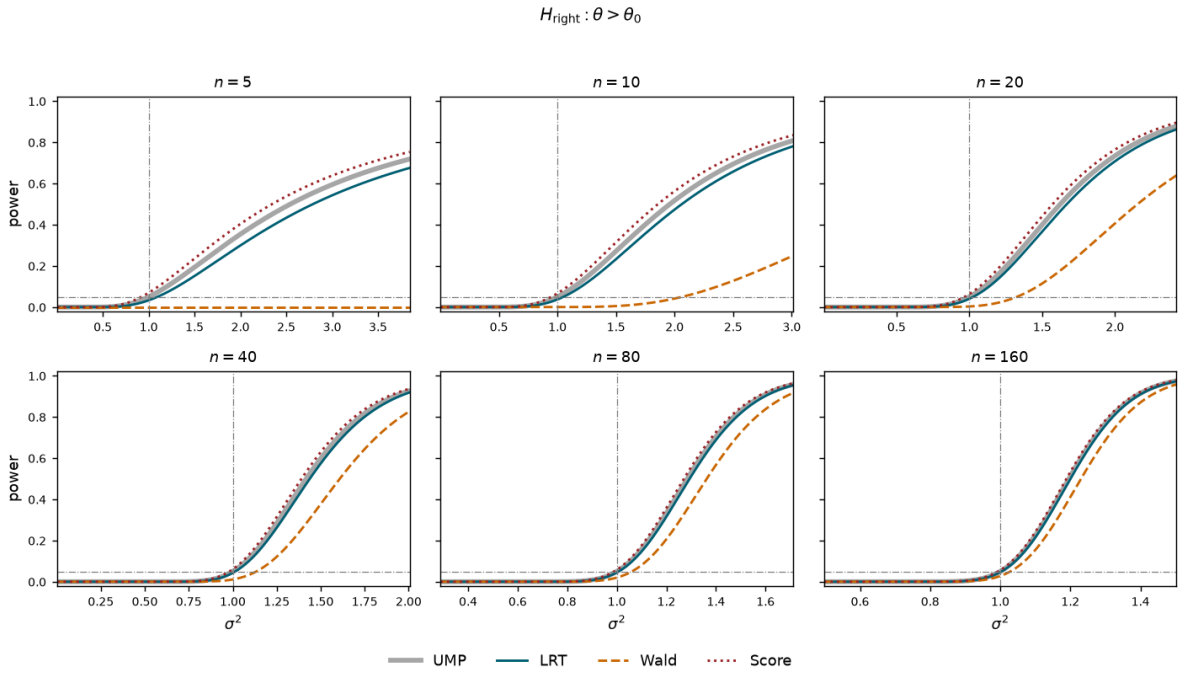


Figure 22: Right alternative ($\sigma_0^2 = 1$).

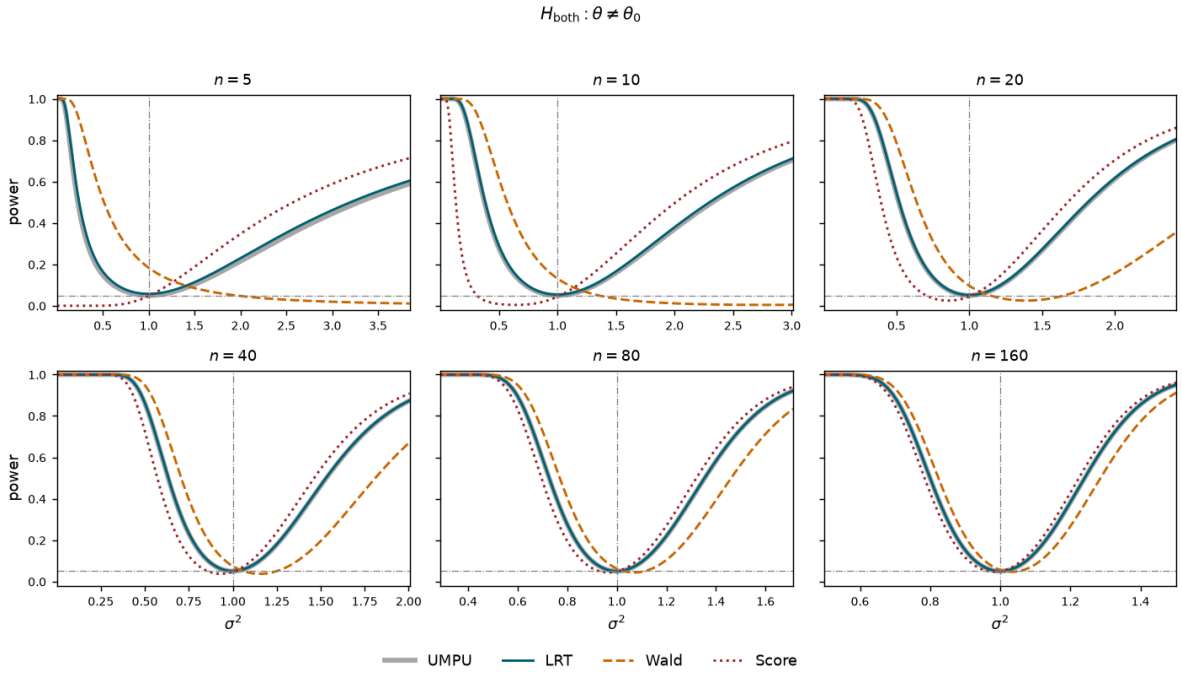


Figure 23: Two-sided alternative ($\sigma_0^2 = 1$).

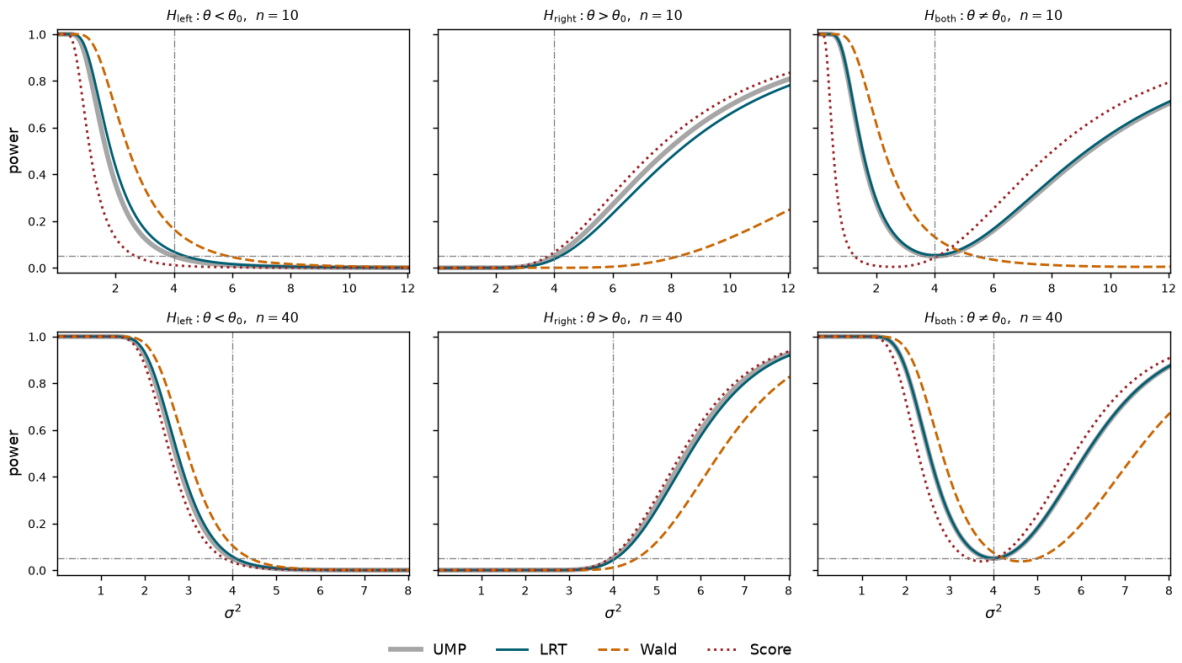


Figure 24: Changed θ_0 ($\sigma_0^2 = 4$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(g) The density is $f_\theta(x) = \theta e^{-\theta x}$, $x > 0$ (θ is the rate). $T = \sum X_i \sim \text{Gamma}(n, \text{rate} = \theta)$, so $2\theta T \sim \chi^2_{2n}$, and $\hat{\theta} = n/T$. Then

$$\ell(\theta) = n \log \theta - \theta T, \qquad U_n(\theta) = n\left(\frac{1}{\theta} - \frac{1}{\hat{\theta}}\right), \qquad I(\theta) = \frac{1}{\theta^2}$$

Test statistics:

$$R_n = \text{sgn}(\hat{\theta} - \theta_0)\sqrt{2n\left[\frac{\theta_0}{\hat{\theta}} - 1 - \log\frac{\theta_0}{\hat{\theta}}\right]}$$

$$W_n = \sqrt{n}\left(1 - \frac{\theta_0}{\hat{\theta}}\right) = S_n$$

So Wald and score are *identical* here.

Optimal tests. $a(\theta) = -\theta$ is *decreasing*, so there is MLR in $-T$. The UMP test rejects H_{right} if $T < \chi^2_{2n,\alpha}/(2\theta_0)$ and H_{left} if $T > \chi^2_{2n,1-\alpha}/(2\theta_0)$. The UMPU test rejects if $2\theta_0 T \notin [w_1, w_2]$, where

$$\int_{w_1}^{w_2} f_{\chi^2_{2n}}(w)\,dw = 1 - \alpha, \qquad \int_{w_1}^{w_2} f_{\chi^2_{2n+2}}(w)\,dw = 1 - \alpha$$

Observations.

- Only two distinct curves appear besides the optimal one. The LRT is closest to the optimal tests (shortfall ≤ 0.03).
- Wald/score sizes are 0.067 (left) and 0.011 (right) at $n = 5$, and 0.054 and 0.045 at $n = 160$. The LRT's at $n = 160$ are 0.047 and 0.053.

- For H_{both}, Wald/score has size 0.044 at $n = 5$, but its power is lopsided: it loses up to 0.33 against the UMPU test at $n = 10$, and still 0.05 at $n = 160$. The reason is that $\sqrt{n}(1 - \theta_0/\hat{\theta})$ is bounded above by $\sqrt{n}$ but unbounded below.
- This is a scale family, so the conclusion does **not** change with θ_0. If we use the mean $1/\theta$ as the parameter, the LRT and optimal tests do not change, but Wald does (it is not invariant to reparametrization).

	H_{left}			H_{right}			H_{both}		
n	LRT	Wald	Score	LRT	Wald	Score	LRT	Wald	Score
main choice: $\theta_0 = 1$				*(second choice $\theta_0 = 5$ gives the same table)*					
5	0.037	0.067	0.067	0.069	0.011	0.011	0.054	0.044	0.044
10	0.040	0.064	0.064	0.062	0.025	0.025	0.052	0.045	0.045
20	0.043	0.061	0.061	0.058	0.034	0.034	0.051	0.047	0.047
40	0.045	0.058	0.058	0.056	0.039	0.039	0.050	0.049	0.049
80	0.046	0.056	0.056	0.054	0.043	0.043	0.050	0.049	0.049
160	0.047	0.054	0.054	0.053	0.045	0.045	0.050	0.050	0.050

Table 7: Exact sizes of the three tests (nominal level 0.05).

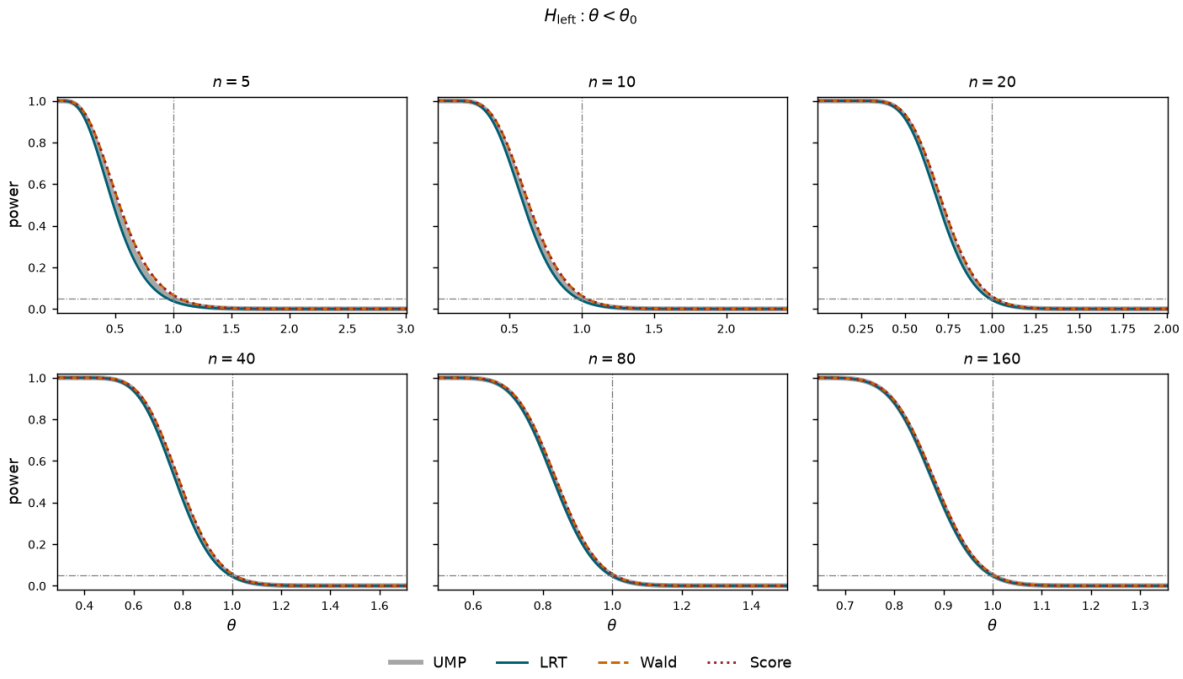


Figure 25: Left alternative ($\theta_0 = 1$).

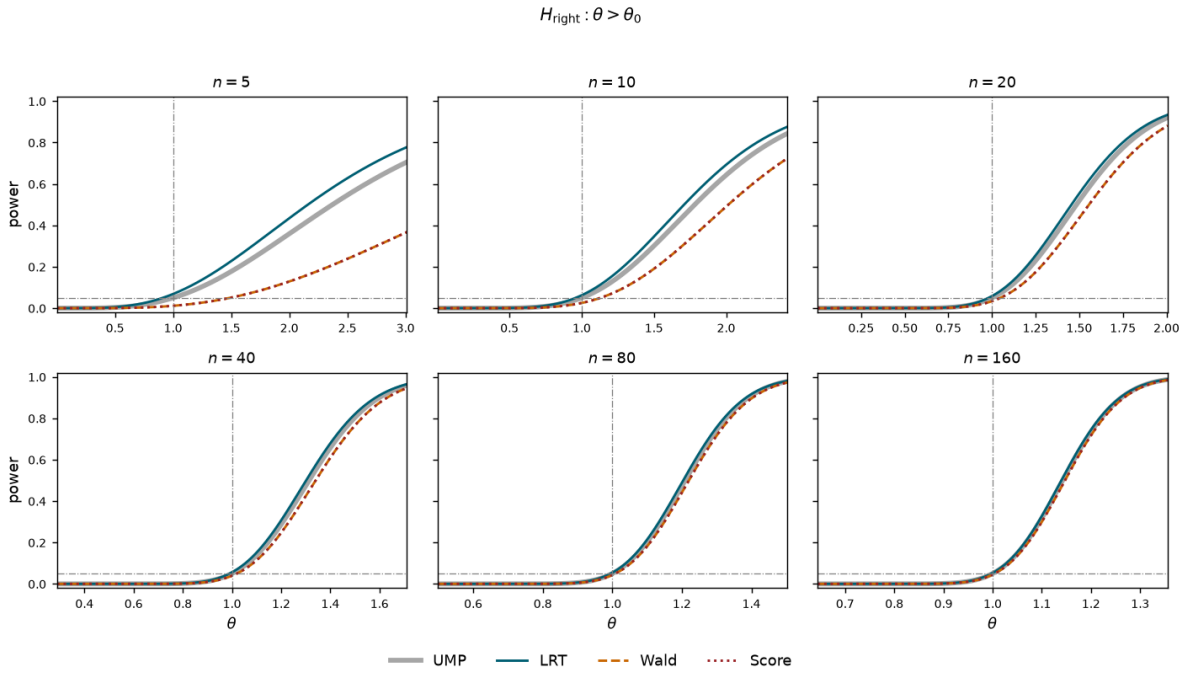


Figure 26: Right alternative ($\theta_0 = 1$).

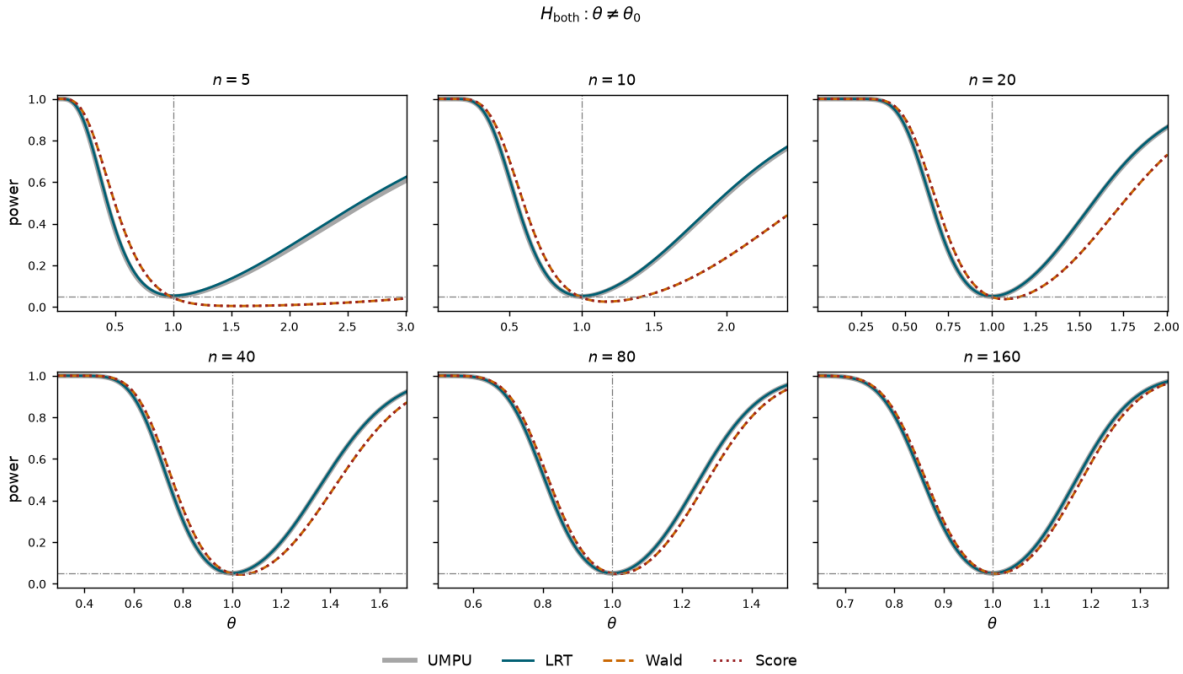


Figure 27: Two-sided alternative ($\theta_0 = 1$).

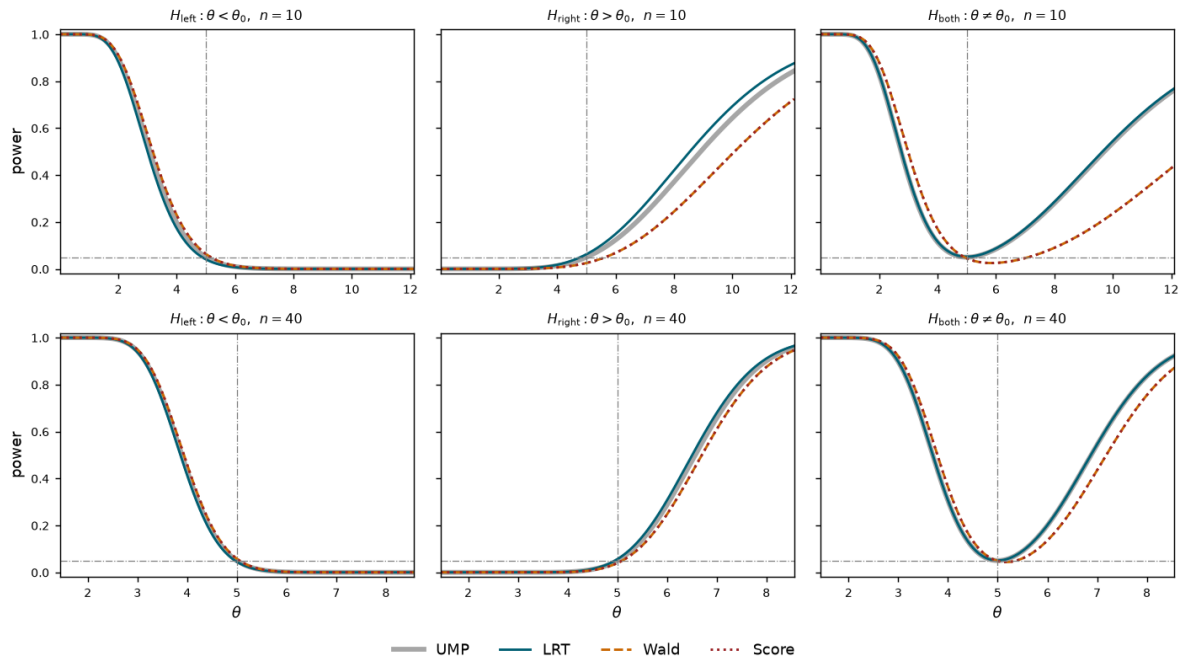


Figure 28: Changed θ_0 ($\theta_0 = 5$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(h) The density is $f_\sigma(x) = \frac{1}{2\sigma} e^{-|x|/\sigma}$. Let $T = \sum |X_i|$; since $|X_i| \sim \text{Exp}(\text{mean } \sigma)$, $2T/\sigma \sim \chi^2_{2n}$. Then

$$\ell(\sigma) = -n \log \sigma - \frac{T}{\sigma} - n \log 2, \qquad U_n(\sigma) = \frac{n(\hat\sigma - \sigma)}{\sigma^2}, \qquad I(\sigma) = \frac{1}{\sigma^2}, \qquad \hat\sigma = \frac{T}{n}$$

Test statistics:

$$R_n = \text{sgn}(\hat\sigma - \sigma_0) \sqrt{2n \left[\frac{\hat\sigma}{\sigma_0} - 1 - \log \frac{\hat\sigma}{\sigma_0} \right]}$$

$$W_n = \sqrt{n} \left(1 - \frac{\sigma_0}{\hat\sigma} \right), \qquad S_n = \sqrt{n} \left(\frac{\hat\sigma}{\sigma_0} - 1 \right)$$

Optimal tests. $a(\sigma) = -1/\sigma$ is increasing, so there is MLR in T. The UMP test rejects H_{right} if $T > \sigma_0 \chi^2_{2n,1-\alpha}/2$ and H_{left} if $T < \sigma_0 \chi^2_{2n,\alpha}/2$. The UMPU test rejects if $2T/\sigma_0 \notin [w_1, w_2]$, with the same w_1, w_2 as in (g).

Observations.

- The pivot is the same as in (g), but the parametrization is reversed, so the roles of Wald and score swap and they no longer coincide. The picture is like (f): the LRT is close to the optimal tests (shortfall ≤ 0.03).
- Wald is liberal on the left (size 0.165 at $n = 5$, 0.068 at $n = 160$) and very conservative on the right (< 0.0005 at $n = 5$, only 0.034 at $n = 160$). Score is the mirror image (0.011 left and 0.067 right at $n = 5$).
- For H_{both} and $n = 10$ the maximum shortfall from the UMPU test is 0.47 (Wald) and 0.33 (score); at $n = 160$ it is still 0.09 and 0.05.
- This is a scale family, so the conclusion does **not** change with σ_0.

n	H_{left} LRT	Wald	Score	H_{right} LRT	Wald	Score	H_{both} LRT	Wald	Score
main choice: $\sigma_0 = 1$ *(second choice* $\sigma_0 = 3$ *gives the same table)*									
5	0.069	0.165	0.011	0.037	0.000	0.067	0.054	0.132	0.044
10	0.062	0.129	0.025	0.040	0.003	0.064	0.052	0.096	0.045
20	0.058	0.105	0.034	0.043	0.011	0.061	0.051	0.074	0.047
40	0.056	0.088	0.039	0.045	0.020	0.058	0.050	0.062	0.049
80	0.054	0.076	0.043	0.046	0.028	0.056	0.050	0.056	0.049
160	0.053	0.068	0.045	0.047	0.034	0.054	0.050	0.053	0.050

Table 8: Exact sizes of the three tests (nominal level 0.05).

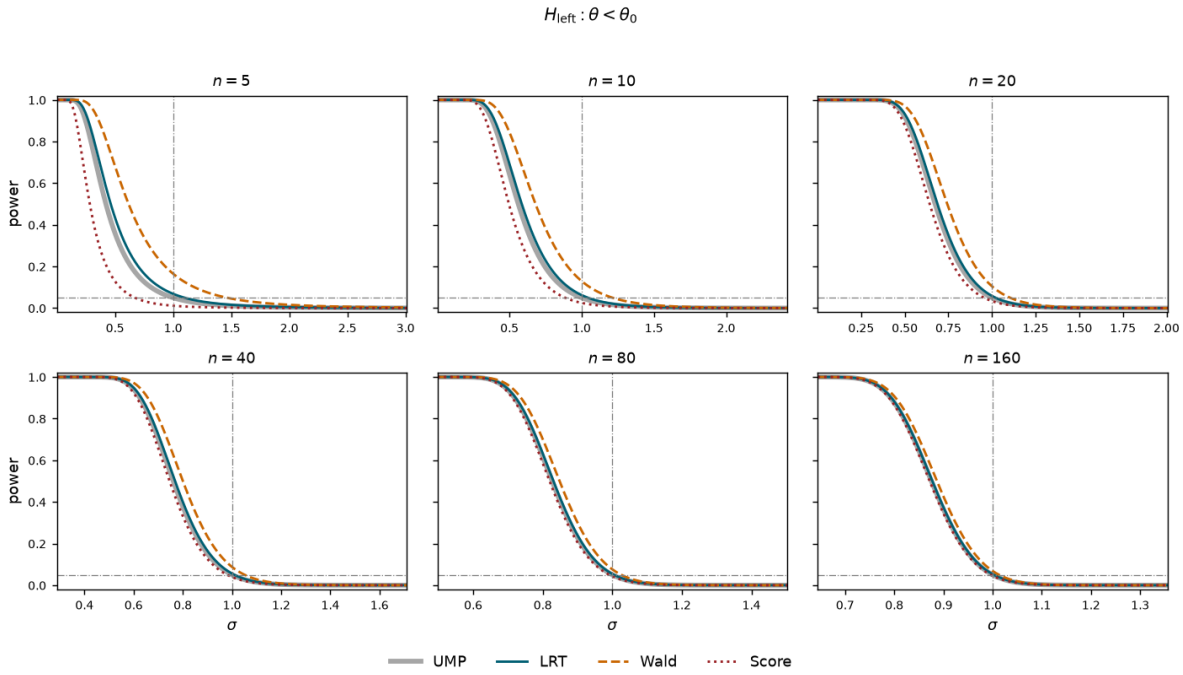


Figure 29: Left alternative ($\sigma_0 = 1$).

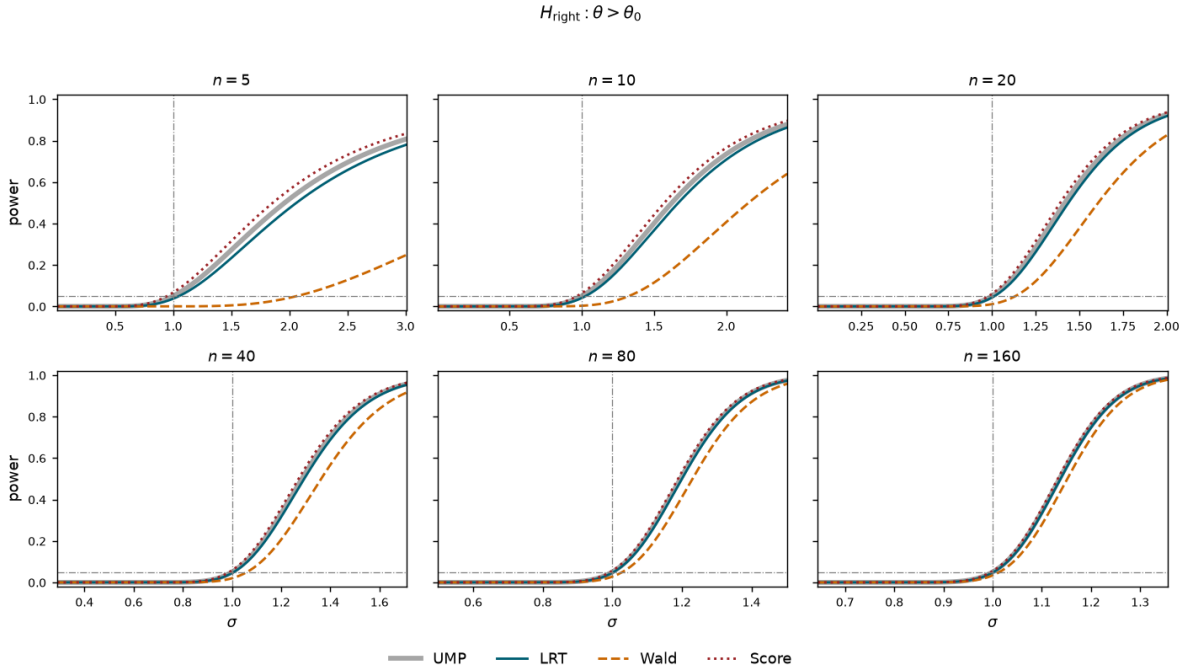


Figure 30: Right alternative ($\sigma_0 = 1$).

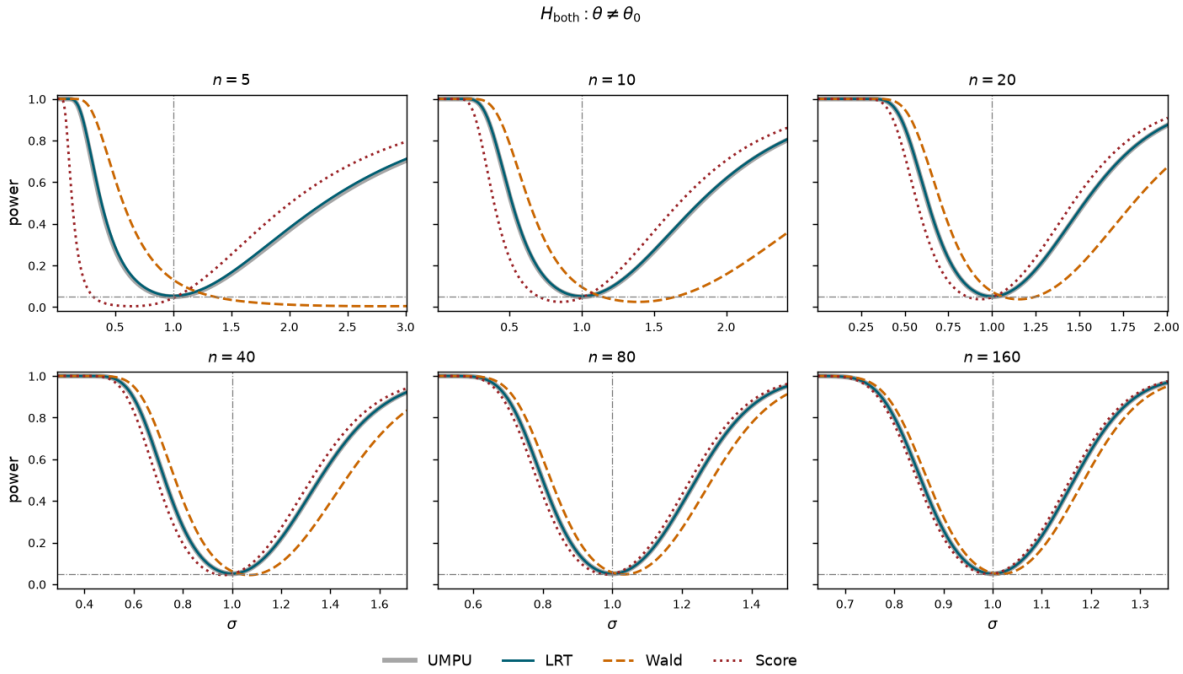


Figure 31: Two-sided alternative ($\sigma_0 = 1$).

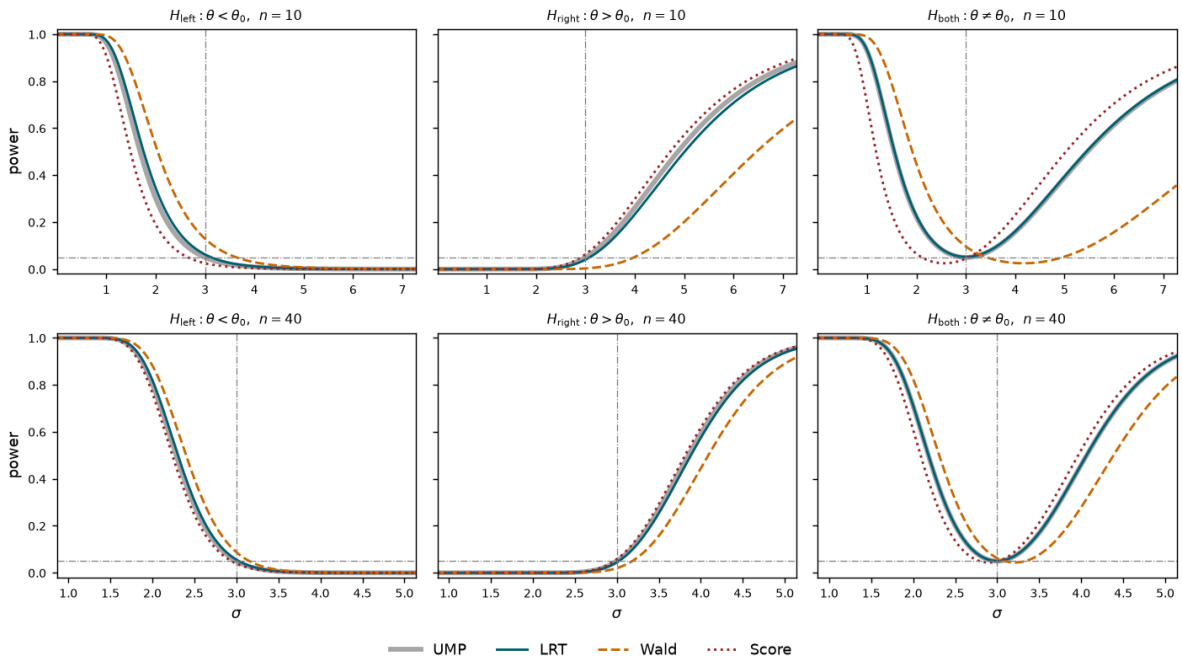


Figure 32: Changed θ_0 ($\sigma_0 = 3$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

(i) The density is

$$f_\theta(x) = (2\pi\theta)^{-1/2} \exp\{-(x-\theta)^2/(2\theta)\} = (2\pi\theta)^{-1/2} e^{-\theta/2} \cdot e^x \cdot \exp\{-x^2/(2\theta)\},$$

a one-parameter exponential family with $a(\theta) = -1/(2\theta)$ and $\tau(x) = x^2$. Let $T = \sum X_i^2$ and $m_2 = T/n$. Then

$$\ell(\theta) = -\frac{n}{2}\log\theta - \frac{nm_2}{2\theta} - \frac{n\theta}{2} + \sum x_i + \text{const}, \qquad U_n(\theta) = \frac{n(m_2 - \theta - \theta^2)}{2\theta^2}$$

Setting $U_n = 0$ gives $\theta^2 + \theta - m_2 = 0$, so

$$\hat\theta = \frac{-1 + \sqrt{1 + 4m_2}}{2}, \qquad m_2 = \hat\theta + \hat\theta^2$$

Since $\partial_\theta^2 \log f_\theta(x) = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3}$ and $E[X^2] = \theta + \theta^2$,

$$I(\theta) = -\frac{1}{2\theta^2} + \frac{\theta + \theta^2}{\theta^3} = \frac{1 + 2\theta}{2\theta^2}$$

Using $\hat\theta^2 + \hat\theta - \theta_0^2 - \theta_0 = (\hat\theta - \theta_0)(\hat\theta + \theta_0 + 1)$, the test statistics are

$$-2\log\Lambda_n = n\left[\log\frac{\theta_0}{\hat\theta} + (\hat\theta + 1)\left(\frac{\hat\theta}{\theta_0} - 1\right) + \theta_0 - \hat\theta\right], \qquad R_n = \operatorname{sgn}(\hat\theta - \theta_0)\sqrt{-2\log\Lambda_n}$$

$$W_n = (\hat\theta - \theta_0)\frac{\sqrt{n(1 + 2\hat\theta)/2}}{\hat\theta}$$

$$S_n = \sqrt{\frac{n}{2}}\,\frac{(\hat\theta - \theta_0)(\hat\theta + \theta_0 + 1)}{\theta_0\sqrt{1 + 2\theta_0}}$$

All three are increasing in $\hat{\theta}$ (equivalently in T), so each rejection region has the form $\{\hat{\theta} < a_L\} \cup \{\hat{\theta} > a_R\}$. We find a_L, a_R numerically by solving $T_n(\hat{\theta}) = \pm z$.

Optimal tests. $a(\theta)$ is increasing, so there is MLR in T. Under θ_0, $T/\theta_0 \sim \chi'^2_n(\lambda)$ (non-central) with $\lambda = n\theta_0$, since $X_i/\sqrt{\theta_0} \sim N(\sqrt{\theta_0}, 1)$. The UMP test rejects H_{right} if $T > \theta_0 q_{1-\alpha}$ and H_{left} if $T < \theta_0 q_\alpha$, where q_u is the u-quantile of $\chi'^2_n(n\theta_0)$. The UMPU test rejects if $T < \theta_0 w_1$ or $T > \theta_0 w_2$, where $W \sim \chi'^2_n(\lambda)$ and

$$P(w_1 < W < w_2) = 1-\alpha, \qquad nP(w_1 < W' < w_2)+\lambda P(w_1 < W'' < w_2) = (1-\alpha)(n+\lambda)$$

with $W' \sim \chi'^2_{n+2}(\lambda)$, $W'' \sim \chi'^2_{n+4}(\lambda)$. (The second equation is $E_{\theta_0}[T\phi] = \alpha E_{\theta_0}[T]$ with $E_{\theta_0}[T] = n\theta_0(1+\theta_0)$, using $w f_{\nu,\lambda}(w) = \nu f_{\nu+2,\lambda}(w) + \lambda f_{\nu+4,\lambda}(w)$.)

Observations.

- For $\theta_0 = 1$ the picture is like the scale families. The LRT is almost the same as the UMP/UMPU tests for $n \geq 20$ (shortfall ≤ 0.03 one-sided, ≤ 0.00 two-sided, $n \geq 10$).
- Wald is liberal on the left (0.154 at $n = 5$, 0.064 at $n = 160$) and conservative on the right (0.007 and 0.038). Score is the mirror image (0.004 and 0.068 at $n = 5$).
- The two-sided Wald and score curves are shifted left and right of the UMPU curve for $n \leq 40$, with maximum shortfall up to 0.19 (Wald) and 0.44 (score) at $n = 10$. At $n = 160$ all three agree with the UMPU test up to 0.05.
- *Changing* θ_0. This is not a scale family, so θ_0 matters, though only mildly. For $\theta_0 = 0.1$ the mean θ is small compared to the standard deviation $\sqrt{\theta}$, so the model behaves like $N(0, \theta)$ in (f), and the sizes are almost the same as in (f) (Wald left: 0.207 at $n = 5$ vs 0.216). For $\theta_0 = 1$ they are a bit better (0.154). The ranking LRT > score $\approx$ Wald does not change.

	H_{left}			H_{right}			H_{both}		
n	LRT	Wald	Score	LRT	Wald	Score	LRT	Wald	Score
main choice: $\theta_0 = 1$									
5	0.074	0.154	0.004	0.037	0.007	0.068	0.058	0.122	0.045
10	0.065	0.118	0.021	0.040	0.014	0.065	0.054	0.088	0.044
20	0.060	0.095	0.032	0.043	0.021	0.061	0.052	0.070	0.046
40	0.057	0.080	0.038	0.045	0.028	0.059	0.051	0.060	0.048
80	0.055	0.070	0.042	0.046	0.034	0.056	0.050	0.055	0.049
160	0.053	0.064	0.045	0.047	0.038	0.055	0.050	0.053	0.050
second choice: $\theta_0 = 0.1$									
5	0.078	0.207	0.000	0.033	0.000	0.070	0.058	0.174	0.048
10	0.068	0.157	0.012	0.037	0.001	0.067	0.054	0.124	0.044
20	0.062	0.123	0.025	0.041	0.007	0.064	0.052	0.091	0.045
40	0.058	0.100	0.034	0.043	0.015	0.060	0.051	0.072	0.047
80	0.056	0.084	0.039	0.045	0.023	0.058	0.050	0.061	0.049
160	0.054	0.074	0.043	0.046	0.030	0.056	0.050	0.056	0.049

Table 9: Exact sizes of the three tests (nominal level 0.05).

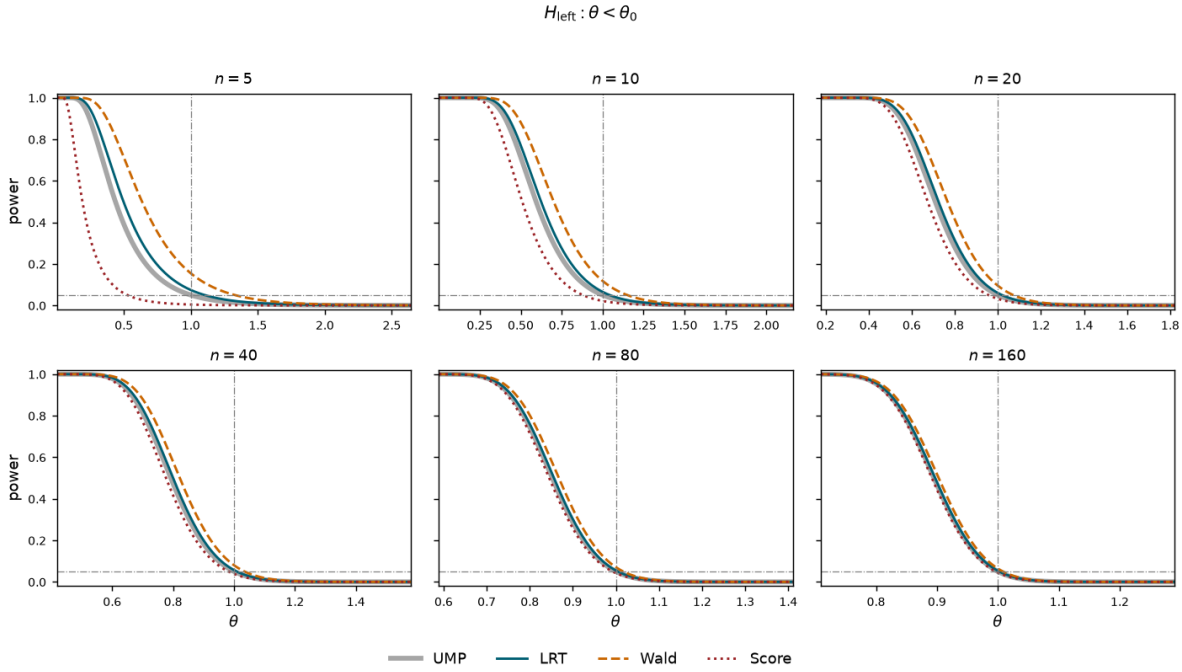


Figure 33: Left alternative ($\theta_0 = 1$).

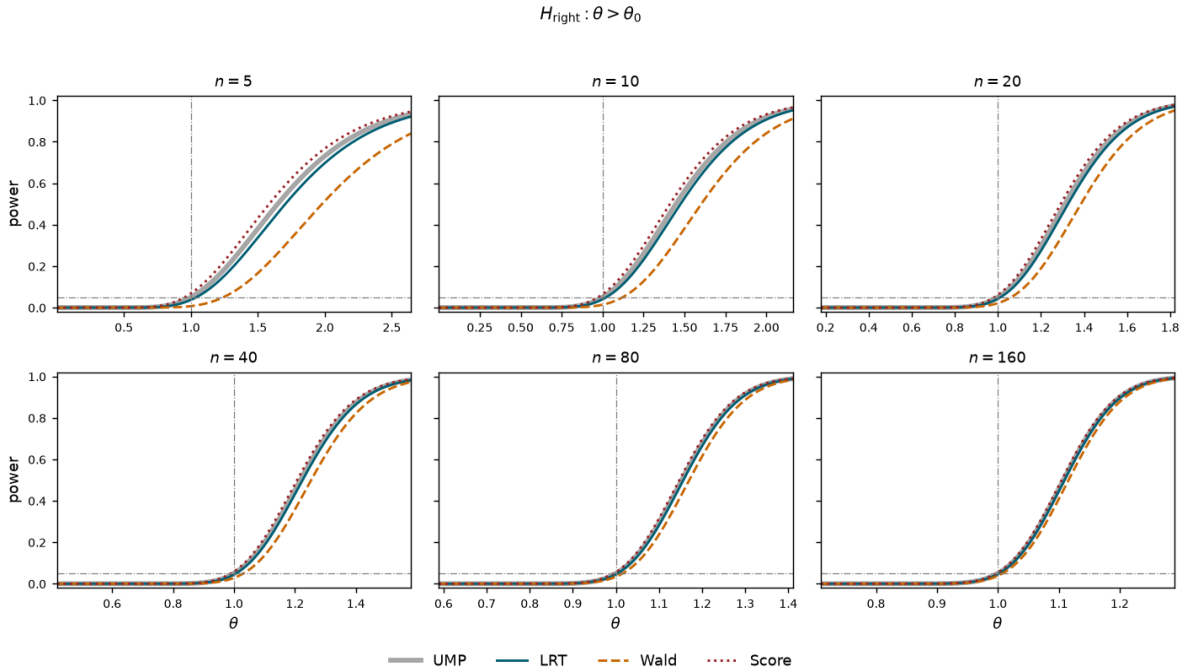


Figure 34: Right alternative ($\theta_0 = 1$).

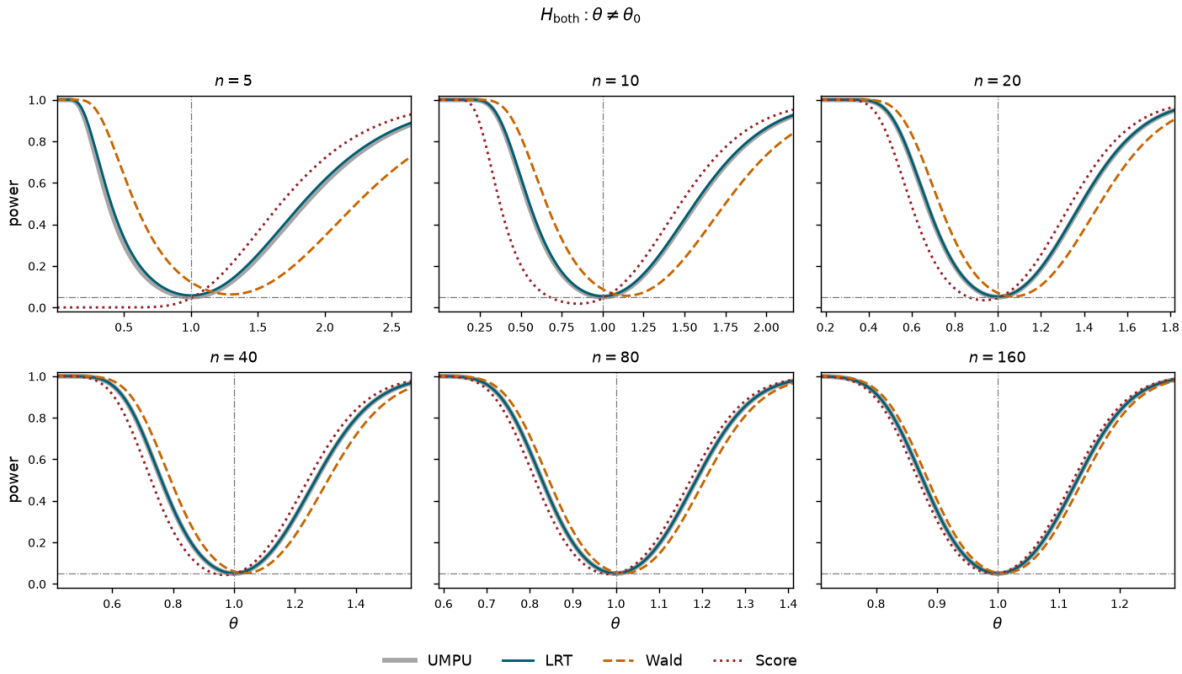


Figure 35: Two-sided alternative ($\theta_0 = 1$).

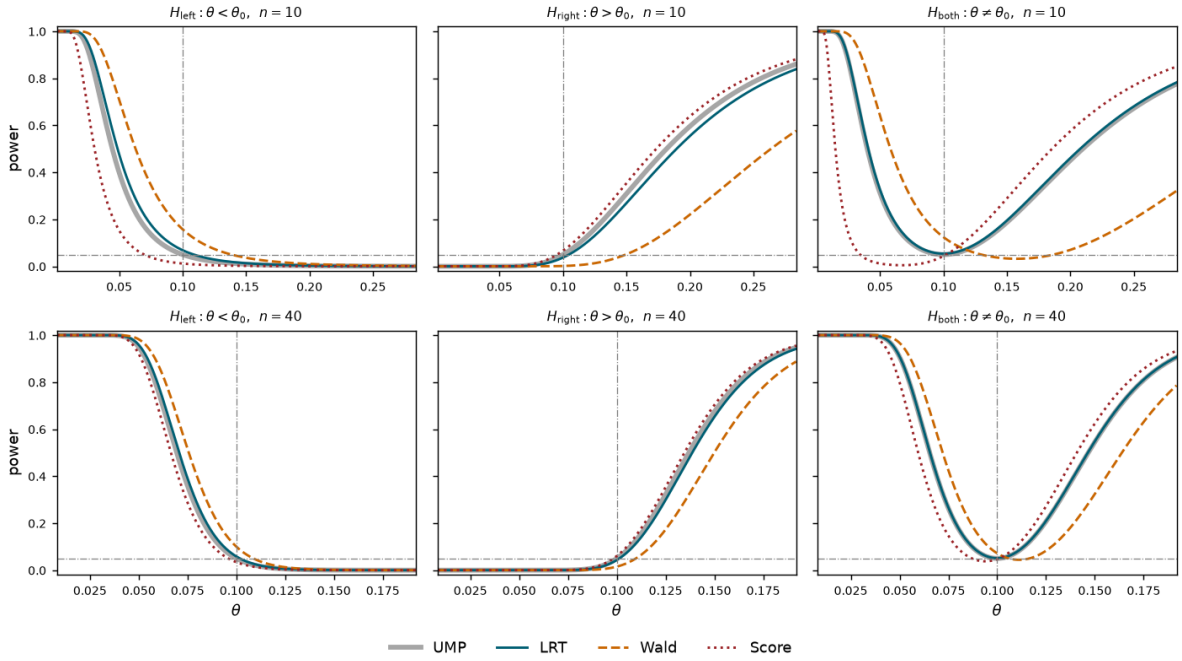


Figure 36: Changed θ_0 ($\theta_0 = 0.1$), $n = 10$ and $n = 40$; columns are $H_{\text{left}}, H_{\text{right}}, H_{\text{both}}$.

Overall comparison

- **LRT.** In all regular models it is the closest to the optimal tests, even for $n = 5$ or 10. It uses the whole log-likelihood instead of a quadratic approximation at one point, and it is invariant under reparametrization. In the uniform model, with

the exact cut-off, it is exactly the UMP/UMPU test (for H_{right} it has size 0). Its demerits: it needs the MLE, and the cut-off sometimes needs correcting, as in (d).

- **Wald.** The worst, or tied for worst, in almost every setup. It is not invariant under reparametrization (e.g. σ vs. σ^2), is unstable when $\hat{\theta}$ is near the boundary, and is liberal in one tail and conservative in the other whenever $\hat{\theta}$ is skewed. It can fail to be a test at all: for $n = 5$ it cannot reject H_{right} in (f). In (d) it fails completely.

- **Score.** It needs only the null model. In skewed models it mirrors Wald: conservative where Wald is liberal, and vice versa. It is often better than Wald but less reliable than the LRT at small n, and it can be unable to reject (parts (c), (f)). In (d) its statistic is constant; in (e) it equals the other two tests; in (g) it equals Wald.

- **Comparison with UMP/UMPU.** The optimal tests have size exactly α for every n (using randomization in the discrete cases); the three classical tests only have size close to α for large n. In the discrete models (a)–(c) their sizes jump around α and settle only for large n; in the continuous models they converge monotonically but slowly (e.g. Wald in (f) at $n = 160$). For $n = 160$ all three agree with the UMP/UMPU tests in the regular models. So the classical tests only cost us at small and moderate n.

- **Effect of θ_0.** The conclusion is unchanged in the location/scale models (d)–(h), where the power depends only on $\theta - \theta_0$ or θ/θ_0. It **does** change in (a), (b), (c), (i): the closer θ_0 is to the boundary (small p_0 or λ_0; p_0 near 1 for the geometric), the more skewed $\hat{\theta}$ is and the larger the n needed for the asymptotic tests to reach size α.